Gaming And Mental Health

Submitted By :

Mohamed Ahmed Bedier

Youssef Talaat AbdElAziz

Shahd Mohamed Sayed

Ebrahim Nasser Elnemr

Belal Ahmed Mohamed

Supervised By :

Dr. Amal Mahmoud

DEPI Round 4

Data Analysis Specialist Track

Acknowledgement

We would like to express our sincere gratitude to everyone who supported us throughout the completion of this project.

First, we would like to thank the DEPI program for providing us with the opportunity to enhance our knowledge and practical skills in Data Analysis through this valuable learning experience.

We would like to express our deepest appreciation to our supervisor, **Dr. Amal Mahmoud**, for her continuous guidance, valuable advice, constructive feedback, and encouragement throughout this project. Her support and expertise played a significant role in helping us successfully complete this work.

We would also like to thank our instructors and mentors for their continuous support and valuable insights during the project.

Finally, we would like to express our heartfelt gratitude to our families, friends, and teammates for their encouragement, patience, and cooperation throughout this journey. Their support was invaluable in achieving this project successfully.

Abstract

The rapid growth of the gaming industry has raised increasing concerns about its potential impact on mental health. This project aims to analyze the relationship between gaming habits and mental well-being using a real-world dataset containing demographic information, gaming behavior, lifestyle factors, and mental health indicators.

The data was cleaned, transformed, and analyzed using Microsoft Excel, SQL, Python, Microsoft Power BI, and Tableau. Various visualizations and dashboards were created to identify trends, patterns, and key performance indicators related to gaming behavior and mental health. The analysis focuses on factors such as gaming hours, sleep duration, stress levels, social interaction, and addiction risk.

The results provide meaningful insights into how different gaming habits may be associated with mental health outcomes. This project demonstrates the practical application of data analysis techniques to support informed decision-making and highlight the importance of maintaining a healthy balance between gaming and overall well-being.

Table of Contents

1. Introduction

1.1 Background

The gaming industry has grown significantly over the past decade, becoming one of the most popular forms of entertainment worldwide. While gaming offers several benefits, such as improving problem-solving skills, creativity, and social interaction, excessive gaming has raised concerns about its potential impact on mental health. Researchers have found that prolonged gaming may be associated with increased stress, anxiety, poor sleep quality, and a higher risk of gaming addiction.

With the growing availability of gaming-related data, data analytics provides an effective way to explore behavioral patterns and identify meaningful relationships between gaming habits and mental well-being.

1.2 Problem Definition

The increasing amount of time spent on video games has raised concerns about its impact on mental health. However, the relationship between gaming behavior and psychological well-being is often complex and difficult to understand without proper data analysis. This project addresses this challenge by analyzing gaming and mental health data to uncover patterns and provide insights that support better awareness and decision-making.

1.3 Motivation

The motivation behind this project is the increasing popularity of video games and the growing importance of mental health in today's society. By applying data analysis techniques, we aim to understand how different gaming habits relate to mental health indicators and present the findings through interactive dashboards that simplify data interpretation.

1.4 Project Objectives

The main objectives of this project are:

- Analyze the Gaming and Mental Health dataset.
- Clean and preprocess the collected data.
- Perform Exploratory Data Analysis (EDA).
- Identify patterns and relationships between gaming habits and mental health.
- Develop interactive dashboards for data visualization.
- Present meaningful insights and recommendations based on the analysis.

1.5 Tools and Technologies

The project was completed using the following tools:

- Microsoft Excel
- SQL
- Python
- Microsoft Power BI
- Tableau

1.6 Report Organization

This report is organized into four chapters. Chapter One introduces the project and its objectives. Chapter Two explains the methodology followed throughout the project. Chapter Three presents the dashboards, data analysis, key insights, and recommendations. Finally, Chapter Four concludes the project and suggests future improvements.

2. Methodology

2.1 Introduction

This chapter describes the methodology followed to analyze the Gaming and Mental Health dataset. The project involved collecting the dataset, cleaning and preprocessing the data, performing exploratory analysis, and developing interactive dashboards to present meaningful insights.

2.2 Dataset Description

The dataset contains demographic information, gaming behavior, lifestyle characteristics, and mental health indicators. It includes variables such as age, gender, gaming platform, daily gaming hours, sleep duration, stress level, social interaction, monthly gaming spending, and addiction risk.

2.3 Data Collection

The dataset was obtained from a publicly available online source and imported into Microsoft Excel for initial inspection and preprocessing before being analyzed using SQL, Python, Power BI, and Tableau.

2.4 Data Preprocessing

To ensure data quality, several preprocessing techniques were applied:

- Removing duplicate records.
- Handling missing values.

- Correcting inconsistent values.
- Standardizing categorical variables.
- Converting data types where necessary.
- Validating the dataset before analysis.

2.5 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to better understand the dataset. Statistical summaries and visualizations were used to identify trends, detect outliers, and examine relationships between different variables.

2.6 SQL Analysis

SQL was used to query the dataset, calculate summary statistics, filter records, group data, and answer key business questions related to gaming behavior and mental health.

2.7 Python Analysis

Python was used for data preprocessing, additional exploratory analysis, and visualization. Libraries such as Pandas, NumPy, and Matplotlib were utilized to manipulate data and support the analytical process.

2.8 Dashboard Development

Interactive dashboards were developed using Microsoft Power BI and Tableau. Various charts, KPIs, filters, and slicers were implemented to allow users to explore the data dynamically and gain meaningful insights.

2.9 Chapter Summary

This chapter described the complete analytical methodology, starting from data collection and preprocessing to dashboard development. The following chapter presents the visual analysis and key findings.

3. Detailed Project Background

This project carries out a comprehensive analysis of a dataset that links video gaming habits with human psychological, behavioral, and physical well-being. The core goal is to surface real evidence from the data on how intensive gaming correlates with physical strain, sleep quality, academic/professional productivity, and potential addiction risk.

To get there in an organized way, we followed a full data-analysis lifecycle: understanding the raw data, cleaning and preprocessing it, building a structured database (Star Schema), extracting KPIs and analysis, and finally presenting everything through interactive dashboards.

3.1 Star Schema Overview

Before going into details, it's worth noting that the data was eventually transformed from its original flat structure into a Star Schema database, made up of one central Fact Table and six Dimension Tables. We explain this fully in the Data Modeling section below.

- Fact_Gaming_Mental_Health: the central table, holding all numeric measures (daily gaming hours, sleep hours, academic performance, social isolation score, etc.) plus the foreign keys linking it to the dimension tables.
- Dim_Platform / Dim_Game / Dim_Player / Dim_Sleep / Dim_Addiction / Dim_PhysicalStatus: dimension tables holding the descriptive attributes for each area of the topic.

4. Challenges in Understanding the Data

The very first and most important step we took, before writing any code, was sitting down and understanding the dataset itself really well. The data contained a lot of interconnected columns (gaming behavior, psychological data, physical data, social and financial data), so we had to understand the meaning of each column individually, and how it related to the rest, before doing any cleaning or analysis.

This mattered a lot, because any wrong assumption at this stage could have led to completely misleading results. For example, two columns, grades_gpa and work_productivity_score, showed a noticeable number of missing (Null) values. If we had rushed and just filled them with zero or dropped them outright, we would have distorted the entire analysis. Instead, we took the time to understand the real reason behind these gaps first — which we explain in detail in the Exploratory Data Analysis (EDA) section.

□ **Key Takeaway**

Understanding the domain context of the data before applying any statistical processing is the most important step in any data analysis project — processing without understanding turns naturally-occurring gaps like these into decisions that harm the entire final result.

5. Environment Setup and Loading the Data

5.1 Importing Libraries

We used a core set of Python libraries for analysis, visualization, and database work: pandas and numpy for data handling, matplotlib and seaborn for plotting, streamlit for building the interactive dashboards, SQLAlchemy for working with the database, and eralchemy2 to automatically generate the ER diagram (ERD).

□ **Code Example:** *Importing the libraries used across the project*

```
import pandas as pd

import numpy as np

# Visualization libraries for plotting charts

import matplotlib.pyplot as plt

import seaborn as sns

# Streamlit for building the web app dashboard

!pip install streamlit -q

import streamlit as st

# SQLAlchemy tools for DB connection and ORM mapping

from sqlalchemy import create_engine, Column, Integer, String, Float, ForeignKey

from sqlalchemy.orm import declarative_base, relationship

# Library to generate and render ER diagrams

from eralchemy2 import render_er

import os
```

5.2 Loading the Dataset

Next, we loaded the CSV file for the dataset into a single DataFrame, which became the foundation for every step that follows.

□ **Code Example:** *Loading the dataset*

```
# Loading the dataset from a specific path

df = pd.read_csv("Gaming and Mental Health.csv")
```

6. Basic Information About the Dataset

Once the data was loaded, the natural next step to understand it better was to look at its basic information: number of rows and columns, data types per column, and a preview of the first and last rows. This step is important because it gives a quick picture of the data's size and quality before any detailed analysis.

- df.describe(): gives a quick statistical summary (mean, standard deviation, min, max) for all numeric columns.
- df.shape: shows the overall number of rows and columns.

- df.info(): shows the data type of each column (numeric, text...) and the count of non-null values — the first signal of missing values.
- df.head() and df.tail(): show the first and last five rows, to make sure the data loaded correctly and looks reasonable.

 Code Example: *Exploring the basic information of the dataset*

Summary statistics for numerical columns (mean, min, max, etc.)

df.describe()

Check the number of rows and columns (dimensions of the dataframe)

print(df.shape)

Display dataframe summary, including data types and missing values

df.info()

Preview the first 5 rows of the dataset

df.head()

Preview the last 5 rows of the dataset

df.tail()

7. Statistics

The statistics section is split into two parts: Descriptive statistics, which describes the shape of the data and flags outliers, and Inferential statistics, which formally tests real relationships and differences between variables using statistical tests.

7.1 Descriptive Statistics

Here we used the 3-Sigma Rule, a statistical rule stating that any value more than 3 standard deviations away from the mean is considered an outlier. We first applied this rule to the daily gaming hours column, then looped it across every numeric column in the dataset at once, to confirm there were no strange or unreasonable values that could affect later analysis.

 Code Example: *Detecting outliers using the 3-Sigma rule*

Calculate mean and standard deviation for gaming hours

mean = df["daily_gaming_hours"].mean()

std = df["daily_gaming_hours"].std()

Define thresholds for outliers using the 3-sigma rule

```
lower_bound = mean - 3 * std

upper_bound = mean + 3 * std

# Filter the dataframe to isolate the outliers

outliers_df = df[(df["daily_gaming_hours"] < lower_bound) | (df["daily_gaming_hours"] > upper_bound)]

print(f"Lower Bound : {lower_bound:.2f}")

print(f"Upper Bound : {upper_bound:.2f}")

print(f"Total Outliers Found : {len(outliers_df)}")

# Apply the same idea to all numerical columns at once

numeric_cols = df.select_dtypes(include=['number']).columns

for col in numeric_cols:

mean = df[col].mean()

std = df[col].std()

lower_bound = mean - 3 * std

upper_bound = mean + 3 * std

outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]

print(f"□ {col}: {len(outliers)} outliers found")
```

7.2 Inferential Statistics

In this part we used four different statistical tests, each answering a specific research question, and each one paired with a chart to help visualize the result, not just read it as numbers.

a) Independent Two-Sample T-Test

We used this test to check whether the difference in average sleep hours between two groups (Low addiction risk vs. Severe addiction risk) is a real, statistically significant difference, or could just be due to chance. The result was shown as a Boxplot comparing the sleep-hours distribution between the two groups, with the p-value printed directly on the chart to make the statistical significance clear at a glance.

□ **Code Example:** *T-Test with a Boxplot to visualize the result*

```
# Separate the two groups (sleep hours for Low and Severe addiction risk levels)

Low_Addiction = df[df['gaming_addiction_risk_level'] == 'Low']['sleep_hours'].dropna()
```

```
Severe_Addiction = df[df['gaming_addiction_risk_level'] == 'Severe']['sleep_hours'].dropna()

# Execute the independent t-test

t_stat, p_val = stats.ttest_ind(Low_Addiction, Severe_Addiction, equal_var=False)

if p_val < 0.05:

print("Statistically significant result: there is a real difference in average sleep hours.")

else:

print("Not significant: the difference might be due to chance.")

# Draw a Boxplot to compare the distribution of sleep hours

df_filtered = df[df['gaming_addiction_risk_level'].isin(['Low', 'Severe'])]

plt.figure(figsize=(8, 6))

sns.boxplot(x='gaming_addiction_risk_level', y='sleep_hours', data=df_filtered, palette='Set2')

plt.title('Comparison of Sleep Hours by Gaming Addiction Risk Level')

plt.text(0.5, df_filtered['sleep_hours'].max()*0.95, f'p-value: {p_val:.4f}',

ha='center', color='red', bbox=dict(facecolor='white', alpha=0.8))

plt.show()
```

b) One-Way ANOVA (Analysis of Variance)

When we needed to compare more than two groups at once (three addiction-risk levels: Low, Moderate, Severe), running multiple T-tests would inflate the statistical error rate, so we used ANOVA instead. It compares the variance between groups to the variance within each group — if the between-group variance is significantly larger, gaming hours genuinely differ by addiction level. This result was also shown with a Boxplot comparing all three groups.

□ **Code Example:** *ANOVA test across three groups*

```
low_risk = df[df['gaming_addiction_risk_level'] == 'Low']['daily_gaming_hours'].dropna()

mod_risk = df[df['gaming_addiction_risk_level'] == 'Moderate']['daily_gaming_hours'].dropna()

severe_risk = df[df['gaming_addiction_risk_level'] == 'Severe']['daily_gaming_hours'].dropna()

# Execute the One-Way ANOVA test

f_stat, p_val = stats.f_oneway(low_risk, mod_risk, severe_risk)

print(f"F-statistic: {f_stat}, p-value: {p_val}")
```

```
if p_val < 0.05:

print("At least one group differs in mean gaming hours from the others.")

else:

print("No statistically significant difference between the groups.")

# Boxplot comparing the three groups

df_filtered = df[df['gaming_addiction_risk_level'].isin(['Low', 'Moderate', 'Severe'])]

plt.figure(figsize=(9, 6))

sns.boxplot(x='gaming_addiction_risk_level', y='daily_gaming_hours',

data=df_filtered, order=['Low', 'Moderate', 'Severe'], palette='Set2')

plt.show()
```

c) Chi-Square Test of Independence

This test is designed for categorical variables, and we used it to check whether there's a real relationship between gaming platform and addiction-risk level, or whether the distribution is essentially random. The result was shown as a stacked percentage Bar Chart illustrating the distribution of addiction-risk levels across each platform, with the p-value printed above the chart.

□ **Code Example:** *Chi-Square test and Stacked Bar Chart*

```
# Create the contingency table between categorical variables

contingency_table = pd.crosstab(df['gaming_platform'], df['gaming_addiction_risk_level'])

# Execute the Chi-Square test of independence

chi2, p_val, dof, expected = stats.chi2_contingency(contingency_table)

if p_val < 0.05:

print("Significant relationship between gaming platform and addiction risk level.")

else:

print("No statistical relationship; the distribution may be random.")

# Convert to percentages and plot as a stacked bar chart

contingency_pct = pd.crosstab(df['gaming_platform'], df['gaming_addiction_risk_level'],
normalize='index') * 100

ax = contingency_pct.plot(kind='bar', stacked=True, figsize=(10, 6), colormap='viridis')
```

```
plt.title('Distribution of Gaming Addiction Risk Levels across Platforms')
plt.legend(title='Addiction Risk Level', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.show()
```

d) Linear Regression (OLS)

Simple correlation only describes the strength of a relationship, but linear regression goes further and builds a predictive model showing how an independent variable (daily gaming hours) can explain or predict a dependent variable (sleep hours). We used the statsmodels library to compute the line equation (y = mx + c) and the R-squared value, and displayed both directly on a Scatter Plot with a regression line, to visualize the direction and strength of the relationship.

□ **Code Example:** *Building and plotting a linear regression model*

```
import statsmodels.api as sm
# Clean the data from missing values
regression_data = df[['daily_gaming_hours', 'sleep_hours']].dropna()
X = regression_data['daily_gaming_hours']
y = regression_data['sleep_hours']
# Calculate model parameters (Slope, Intercept, R-squared)
X_with_const = sm.add_constant(X)
model = sm.OLS(y, X_with_const).fit()
intercept = model.params['const']
slope = model.params['daily_gaming_hours']
r_squared = model.rsquared
stats_text = f"Equation: y = {slope:.2f}x + {intercept:.2f}\nR²: {r_squared:.3f}"
# Scatter plot with regression line
plt.figure(figsize=(10, 6))
sns.regplot(x='daily_gaming_hours', y='sleep_hours', data=regression_data,
scatter_kws={'alpha': 0.4, 'color': '#1f77b4'},
line_kws={'color': '#d62728', 'linewidth': 2.5})
plt.text(regression_data['daily_gaming_hours'].max()*0.65,
```

```
regression_data['sleep_hours'].max()*0.85, stats_text, fontweight='bold')

plt.show()
```

8. Exploratory Data Analysis (EDA)

After the statistics work, we went back a step to check data quality directly: are there duplicate rows? Are there missing values, and if so, is their proportion normal or does it need to be fixed?

8.1 Checking for Duplicates

We confirmed that every record (record_id) is unique and not duplicated, to make sure each row represents exactly one real person, with no artificial inflation of the sample.

 Code Example: *Checking for duplicate rows*

```
# Check for duplicate entries based on the unique record identifier

duplicates = df["record_id"].duplicated().sum()

print(f"Duplicates: {duplicates}")
```

The result was zero duplicates, meaning the dataset is clean from a duplication standpoint, and every record represents a different person.

8.2 Checking Null Values and Their Proportion

When we ran df.isnull().sum(), we found missing values concentrated almost entirely in two columns: grades_gpa and work_productivity_score. At first glance this could look like a data-quality problem, but once we calculated the proportion and understood the context, it made complete sense: not everyone in the sample is both a student and an employee at the same time, so a student naturally won't have a work_productivity_score, and a worker naturally won't have a grades_gpa. In other words, the gaps weren't a data-collection error, but a natural reflection of each person's category (student / worker).

To be fully certain this interpretation was correct and not just an assumption on our part, we reached out to the original dataset owner and asked directly about the reason for these gaps. They confirmed that our interpretation was correct, and that these gaps were expected and natural given how the data was collected, not the result of an error. That decision saved us a lot of time that could have otherwise gone into trying to "fill in" values that were meant to be empty in the first place.

 Code Example: *Checking null values across all columns*

```
# Count missing (null) values in each column

df.isnull().sum()
```

□ Key Takeaway

Not every Null value is a problem that needs fixing. Sometimes the gap itself carries information (here: whether the person is a student or a worker), and the most important step was confirming that context by going back to the data owner rather than assuming.

9. Data Preprocessing

9.1 Sleep States (sleep_states)

Overview

Sleep duration is an important health indicator. Players were classified into three sleep categories based on their average daily sleep hours to simplify later analysis.

Why was this feature created?

Grouping sleep hours into categories makes comparisons between players easier and improves dashboard readability.

Classification

- Poor: ≤ 4 Hours
- Healthy: >4 and ≤8 Hours
- Over Sleep: >8 Hours

SQL Implementation

```sql

-- sleep_states: bucket each player's sleep_hours into a simple health label
ALTER TABLE Gaming_Mental_Health
ADD COLUMN sleep_states VARCHAR(25);

UPDATE Gaming_Mental_Health
SET sleep_states = CASE
    WHEN sleep_hours  <= 4 then 'Poor'        -- 4 hrs or less -> sleep deprived
    WHEN sleep_hours  <= 8 then 'Healthy'     -- 4-8 hrs -> normal range
    ELSE 'Over_Sleep'                         -- more than 8 hrs
END;

-- Output distribution: Healthy 764 | Poor 147 | Over_Sleep 89
SELECT sleep_states
FROM Gaming_Mental_Health;

```

Expected Outcome

A new categorical column named sleep_states is created, assigning each player to the appropriate sleep category.

9.2 Total Spent (Total_Spent)

Overview

To estimate each player's lifetime gaming expenditure, a new feature called Total Spent was calculated.

Why was this feature created?

Monthly spending alone does not represent a player's total investment. This feature provides a more realistic financial indicator.

Classification

- Formula: Monthly Spending × Years Gaming × 12

SQL Implementation

```
-- Total_spent: lifetime spend estimate = monthly spend x 12 months x years gaming
ALTER TABLE Gaming_Mental_Health
ADD COLUMN Total_spent INT(20);

UPDATE Gaming_Mental_Health
SET Total_spent =monthly_game_spending_usd * years_gaming * 12;

select monthly_game_spending_usd,Total_spent
from Gaming_Mental_Health;
```

Expected Outcome

A numerical column containing the estimated lifetime spending for every player.

9.3 Spend Category (Spend_Category)

Overview

Players were grouped into spending categories using the dataset mean and standard deviation.

Why was this feature created?

This statistical approach adapts to the actual data distribution and avoids arbitrary thresholds.

Classification

- Low
- Mid

- High
- Very High

SQL Implementation

```sql
-- Store the mean and population standard deviation of Total_spent in
-- session variables so they can be reused in the next UPDATE.
-- (MySQL's STDDEV() is the population stddev, not the sample one.)
-- Output: @avg_spent ≈ 7204.60 | @std_spent ≈ 10381.57
SELECT
    AVG(Total_Spent),
    STDDEV(Total_Spent)
INTO @avg_spent, @std_spent
FROM gaming_mental_health;

-- Spend_Category: bucket players into spend tiers using mean/std-dev
-- thresholds (an outlier-aware alternative to plain quartiles)
ALTER TABLE Gaming_Mental_Health
ADD COLUMN Spend_Category VARCHAR(20);

UPDATE Gaming_Mental_Health
SET Spend_Category = CASE
    WHEN Total_Spent <= @avg_spent THEN 'Low'                         -- at/below average
    WHEN Total_Spent <= (@avg_spent + @std_spent) THEN 'Mid'           -- within 1 std-dev above average
    WHEN Total_Spent <= (@avg_spent + 2 * @std_spent) THEN 'High'      -- within 2 std-dev above average
    ELSE 'Very High'                                                   -- more than 2 std-dev above average
END;
```

Expected Outcome

A categorical spending level is assigned to every player.

9.4 Age Group (age_group)

Overview

Players were grouped into age categories for demographic analysis.

Why was this feature created?

Age groups make comparisons easier across dashboards and statistical analyses.

Classification

- Teenager (≤18)
- Young Adult (19–28)
- Adult (>28)

SQL Implementation

```sql

-- age_group: bucket players into age brackets for demographic analysis
ALTER TABLE gaming_mental_health
ADD COLUMN age_group VARCHAR(50);

UPDATE gaming_mental_health
SET age_group = CASE
    WHEN age BETWEEN 12 AND 18 THEN 'Teenager'
    WHEN age BETWEEN 19 AND 28 THEN 'Young Adult'
    ELSE 'Adult'
END;

-- Output distribution: Young Adult 624 | Teenager 326 | Adult 50
SELECT age, age_group FROM gaming_mental_health;

```

Expected Outcome

Each player belongs to a single age category.

9.5 Educational State

Overview

Participants were classified according to their educational and employment status.

Why was this feature created?

This simplifies comparisons between students, workers, and working students.

Classification

- Student
- Worker
- Working Student
- Unknown

SQL Implementation

```sql
-- Educational_State: classify each player by whether they report grades,
-- work productivity, or both — a proxy for "student vs worker vs both"
ALTER TABLE gaming_mental_health
ADD COLUMN Educational_State VARCHAR(20);


UPDATE gaming_mental_health
SET Educational_State = CASE
    WHEN grades_gpa is not null and work_productivity_score is null  then 'Student' -- gpa only
    WHEN grades_gpa is null and work_productivity_score is not null  then 'Worker'  -- work productivity score only
    WHEN grades_gpa is not null and work_productivity_score is not null  then 'Working_Student' -- bothe gpa and work productivity score
    ELSE 'Unknown' -- no gpa and work productivity score
END;


-- Output distribution: Working_Student 428 | Student 326 | Worker 246
SELECT grades_gpa, work_productivity_score,Educational_State FROM gaming_mental_health;
```

Expected Outcome

Each participant receives one educational status.

9.6 Physical Pain

Overview

Eye strain and back/neck pain were combined into a single physical risk indicator.

Why was this feature created?

A single feature is easier to analyze than multiple Boolean variables.

Classification

- High Risk
- Moderate
- No Risk

SQL Implementation

```sql

-- Physical_Pain: combine the two physical-symptom flags into one risk label
ALTER TABLE gaming_mental_health
ADD COLUMN Physical_Pain VARCHAR(20);

UPDATE gaming_mental_health
SET Physical_Pain = CASE
    WHEN eye_strain = 'TRUE' and back_neck_pain = 'TRUE'  then 'High_Risk'   -- both symptoms present
    WHEN (eye_strain = 'False' and back_neck_pain = 'TRUE') OR ( eye_strain ='TRUE' and back_neck_pain = 'False') then 'Moderate'
    ELSE 'NO_Risk'   -- neither symptom present
END;

-- Output distribution: NO_Risk 391 | Moderate 373 | High_Risk 236
SELECT eye_strain, back_neck_pain,Physical_Pain FROM gaming_mental_health;
```

Expected Outcome

Each player receives an overall physical pain level.

9.7 Gaming Hours Category

Overview

Daily gaming hours were categorized using the dataset mean and standard deviation.

Why was this feature created?

Dynamic statistical thresholds provide more meaningful gaming intensity groups.

Classification

- Low
- Mid
- High
- Very High

SQL Implementation

```sql
-- Gaming_Hours_Category: bucket players by daily gaming hours, same
-- mean/std-dev approach used for Spend_Category above
ALTER TABLE gaming_mental_health
ADD COLUMN Gaming_Hours_Category VARCHAR(20);

-- Output: @avg_hours ≈ 6.15 | @std_hours ≈ 2.87
SELECT
    AVG(daily_gaming_hours),
    STDDEV(daily_gaming_hours)
INTO @avg_hours, @std_hours
FROM gaming_mental_health;


UPDATE gaming_mental_health
SET Gaming_Hours_Category = CASE
    WHEN daily_gaming_hours <= @avg_hours THEN 'Low'
    WHEN daily_gaming_hours <= (@avg_hours + @std_hours) THEN 'Mid'
    WHEN daily_gaming_hours <= (@avg_hours + 2 * @std_hours) THEN 'High'
    ELSE 'Very High'
END;

-- Output distribution: Low 529 | Mid 312 | High 126 | Very High 33
SELECT daily_gaming_hours, Gaming_Hours_Category FROM gaming_mental_health;
```

Expected Outcome

Each player is assigned a gaming intensity category.

10. Data Modeling

Once the data was fully preprocessed, the next step was building an organized relational database instead of leaving everything as one flat file. We used SQLite3 to create an in-memory database, loaded the DataFrame as a staging table, and then built the dimension tables and the fact table from it using direct SQL commands.

Each dimension table has a primary key and unique values pulled from the original data, which reduces duplication and improves organization significantly — this is the Star Schema idea introduced earlier.

□ **Code Example:** *Creating the database and dimension tables with SQLite3*

```
import sqlite3

# Create an in-memory SQLite database connection

conn = sqlite3.connect(":memory:")

# Write the main dataframe to a staging table in SQLite

df.to_sql("staging_df", conn, index=False, if_exists="replace")

# -- Dim Platform --

conn.execute("""

CREATE TABLE dim_platform (

platform_id INTEGER PRIMARY KEY AUTOINCREMENT,

gaming_platform TEXT UNIQUE

);

""")

conn.execute("""

INSERT INTO dim_platform (gaming_platform)

SELECT DISTINCT gaming_platform FROM staging_df WHERE gaming_platform IS NOT NULL;

""")

# -- Dim Game --

conn.execute("""

CREATE TABLE dim_game (
```

```
game_id INTEGER PRIMARY KEY AUTOINCREMENT,

game_genre TEXT,

primary_game TEXT,

UNIQUE(game_genre, primary_game)

);

""")

conn.execute("""

INSERT INTO dim_game (game_genre, primary_game)

SELECT DISTINCT game_genre, primary_game FROM staging_df;

""")

# ... the remaining dimension tables were created the same way:

# dim_sleep, dim_addiction, dim_physical, dim_player

# followed by the fact table (fact_gaming_mental_health), which links

# all dimension tables through their foreign keys along with the numeric measures.
```

11. Data Modeling Diagram (ERD)

After building the tables, we wanted to visually confirm that the relationships between them were set up correctly, so we used the eralchemy2 library together with Graphviz to automatically generate an ERD from the SQLAlchemy table definitions. This step matters a lot, since it visually shows the Star Schema shape: the fact table in the middle with the six dimension tables branching out from it, making it easy to review the design and confirm every foreign key correctly links to its primary key.

□ **Code Example:** *Defining tables with SQLAlchemy and rendering the ERD*

```
from sqlalchemy import create_engine, Column, Integer, String, Float, ForeignKey

from sqlalchemy.orm import declarative_base, relationship

from eralchemy2 import render_er

Base = declarative_base()

class DimPlatform(Base):

__tablename__ = 'dim_platform'
```

```
platform_id = Column(Integer, primary_key=True, autoincrement=True)

gaming_platform = Column(String, unique=True)

facts = relationship("FactGamingMentalHealth", back_populates="platform")

class DimGame(Base):

__tablename__ = 'dim_game'

game_id = Column(Integer, primary_key=True, autoincrement=True)

game_genre = Column(String)

primary_game = Column(String)

facts = relationship("FactGamingMentalHealth", back_populates="game")

# ... remaining table definitions (DimSleep, DimAddiction, DimPhysical, DimPlayer,
FactGamingMentalHealth)

# Automatically generate and render the ERD as an image

render_er(Base, "star_schema.png")

# Display the resulting image

from IPython.display import Image, display

display(Image("star_schema.png"))
```

The result was a clear diagram showing that Fact_Gaming_Mental_Health sits at the center of the network, connected to six dimension tables around it (platform, game, player, sleep, addiction, physical status) — the typical shape of any Star Schema in data warehousing.

12. Key Performance Indicators (KPIs)

After completing the modeling, we extracted a set of overall KPIs that give a quick, comprehensive view of the whole sample before diving into more detailed analysis. These KPIs cover total number of players, average daily gaming hours, average sleep hours, average social isolation score, average monthly spending, and total overall spending.

□ **Code Example:** *Calculating the key performance indicators*

KPI	Des cript ion

```
kpis = pd.DataFrame({

"total_players": [len(fact_table)],

"avg_daily_gaming_hours":
```

KPI	Des cript ion
[fact_table["daily_gaming_hours"].mean().round(1)], "avg_sleep_hours": [fact_table["sleep_hours"].mean().round(1)], "avg_social_isolation": [fact_table["social_isolation_score"].mean().round(1)], "avg_monthly_spending": [fa ct_table["monthly_game_spending_usd"].mean().round(1)], "total_spending": [fact_table["total_spent"].sum().round(1)] }) kpis	
total_players	Total n umber of pl ayers in the s ample
avg_daily_gaming_hours	Av erage daily g aming hours
avg_sleep_hours	Av erage sleep hours
avg_social_isolation	Av erage s ocial isol ation score
avg_monthly_spending	Av erage mo nthly spe nding in USD
total_spending	Total spe nding a cross the whole s ample

13. Data Analysis

In this stage we joined the fact table with the various dimension tables using Merge and Groupby operations, to answer specific analytical questions. Each analysis was followed by a chart to visualize the result.

a) Does addiction affect spending and social isolation?

We grouped the data by each addiction level (Low/Moderate/Severe) and computed average spending and average social isolation score for each level, showing the result in two separate Bar Charts.

 Code Example: *Analyzing addiction's effect on spending and isolation*

addiction_analysis = (

fact_table.merge(dim_addiction, on="addiction_id", how="inner")

```
.groupby("gaming_addiction_risk_level")
.agg(
players=("record_id", "count"),
avg_gaming_hours=("daily_gaming_hours", "mean"),
avg_social_isolation=("social_isolation_score", "mean"),
avg_spending=("monthly_game_spending_usd", "mean")
)
.reset_index()
)
plt.figure(figsize=(8, 5))
sns.barplot(data=addiction_analysis, x="gaming_addiction_risk_level", y="avg_spending")
plt.title("Does addiction affect spending?")
plt.show()
plt.figure(figsize=(8, 5))
sns.barplot(data=addiction_analysis, x="gaming_addiction_risk_level",
y="avg_social_isolation")
plt.title("Does addiction affect isolation score?")
plt.show()
```

b) The relationship between sleep, exercise, and gaming hours

We joined the fact table with the sleep dimension table and computed average exercise hours by sleep state, as well as average daily gaming hours by sleep quality, to see whether heavy gaming affects sleep quality and exercise commitment.

□ **Code Example:** *Analyzing the relationship between sleep, exercise, and gaming hours*

```
gaming_sleep_analysis = (
fact_table.merge(dim_sleep, on="sleep_id", how="inner")
.groupby("sleep_state_y")
.agg(players=("record_id", "count"),
avg_sleep_hours=("sleep_hours", "mean"),
```

```
weight_change_kg=("weight_change_kg", "mean"),
avg_exercise_hours=("exercise_hours_weekly", "mean"))
.reset_index()
)
sns.barplot(data=gaming_sleep_analysis, x="sleep_state_y", y="avg_exercise_hours")
plt.title("Does sleep state affect exercise hours?")
plt.show()
gaming_sleep_analysis2 = (
fact_table.merge(dim_sleep, on="sleep_id", how="inner")
.groupby("sleep_quality")
.agg(avg_daily_hours=("daily_gaming_hours", "mean"))
.reset_index()
)
sns.barplot(data=gaming_sleep_analysis2, x="sleep_quality", y="avg_daily_hours")
plt.title("Does gaming hours affect sleep quality?")
plt.show()
```

c) Most-played platform and age group

We joined the fact table with the platform dimension table and the player dimension table, showing the result as two Pie Charts illustrating the distribution of total gaming hours by platform and by age group.

 Code Example: *Analyzing gaming distribution by platform and age group*

```
platform_analysis = (
fact_table.merge(dim_platform, on="platform_id", how="inner")
.groupby("gaming_platform")
.agg(players=("record_id", "count"),
sum_gaming_hours=("daily_gaming_hours", "sum"),
avg_social_isolation=("social_isolation_score", "mean"),
```

```
avg_spending=("monthly_game_spending_usd", "mean"))

.reset_index()

)

plt.pie(x=platform_analysis["sum_gaming_hours"],
labels=platform_analysis["gaming_platform"], autopct='%1.1f%%')

plt.title("Which platform is most played on?")

plt.show()

age_analysis = (

fact_table.merge(dim_player, on="record_id", how="inner")

.groupby("age_group")

.agg(players=("record_id", "count"),

sum_gaming_hours=("daily_gaming_hours", "mean"),

avg_sleep_hours=("sleep_hours", "mean"),

avg_social_isolation=("social_isolation_score", "mean"))

.reset_index()

)

plt.pie(x=age_analysis["sum_gaming_hours"], labels=age_analysis["age_group"],
autopct='%1.1f%%')

plt.title("Which age group is playing the most?")

plt.show()
```

d) The relationship between game genre and spending

We grouped the analysis by game genre to find out which genres take up the most time and spending, sorted the result in descending order, and displayed it as a horizontal Bar Chart.

 Code Example: *Analyzing the relationship between game genre and spending*

```
genre_addiction = (

fact_table.merge(dim_game, on="game_id", how="inner")

.groupby("game_genre")

.agg(players=("record_id", "count"),
```

```
sum_gaming_hours=("daily_gaming_hours", "mean"),

avg_sleep_hours=("sleep_hours", "mean"),

avg_social_isolation=("social_isolation_score", "mean"),

total_spending=("monthly_game_spending_usd", "sum"))

.reset_index()

)

sorted_genres = genre_addiction.sort_values(by="total_spending", ascending=False)

sns.barplot(data=sorted_genres, x="total_spending", y="game_genre")

plt.show()
```

14. Interactive Dashboards

The final stage of the project turned all the previous analysis into three interactive dashboards built with Streamlit and Plotly, using an elegant dark theme with neon accents for a professional look, plus interactive KPI cards that update based on filters the user selects from the sidebar.

1) Player Dashboard

This dashboard focuses on general player data: most-used games and platforms, age and gender distribution, and different gaming patterns. It includes KPI cards at the top and several charts (Bar, Pie, Line) that respond to the sidebar filters.

□ **Code Example:** *Setting up the page and styling the Player Dashboard*

```
import streamlit as st

import plotly.express as px

# Basic page configuration, full-width layout

st.set_page_config(page_title="Player DashBoard", layout="wide")

# Custom CSS for the dark background and neon-styled KPI cards

st.markdown("""

<style>

.stApp { background-color: #0f1123; }

[data-testid="stSidebar"] { background-color: #1b1e3d !important; }
```

```
h1, h2, h3, p, label { color: #FFFFFF !important; font-weight: bold !important; }
.kpi-card {
background-color: #1b1e3d; border-radius: 15px; padding: 15px;
text-align: center; border: 1px solid #3b3f6c;
box-shadow: 0 4px 15px rgba(0,0,0,0.4);
}
.kpi-value { color: #FFFFFF !important; font-size: 24px !important; }
</style>
""", unsafe_allow_html=True)
# ... after this, the sidebar filters are built, KPI cards computed,
# and the charts (Bar / Pie / Line) rendered using Plotly Express
```

2) Addiction Dashboard

This dashboard is dedicated to analyzing addiction risk levels and their relationship with social life, mood state, and spending. It shows how addiction level affects time spent on face-to-face interaction, as well as the relationship between mood state, gaming hours, and spending.

□ **Code Example:** *Addiction Dashboard charts for social life and mood analysis*

```
# Comparing social isolation score with face-to-face interaction by addiction level
df_social_life = df_filtered.groupby('gaming_addiction_risk').agg(
{'Social_Score': 'mean', 'F2F_Social_Hours': 'mean'}
).reset_index()
df_melted = df_social_life.melt(
id_vars='gaming_addiction_risk',
value_vars=['Social_Score', 'F2F_Social_Hours'],
var_name='Metric', value_name='Value'
)
fig = px.bar(df_melted, x='gaming_addiction_risk', y='Value', color='Metric', barmode='group')
fig.update_layout(title="How Addiction Risk Affects Social Life")
```

```
st.plotly_chart(fig, use_container_width=True)

# Relationship between mood state and total spending

fig = px.line(df_spend, x='Mood_State', y='Spend_Value', markers=True)

fig.update_layout(title="Total Spending by Mood State")

st.plotly_chart(fig, use_container_width=True)
```

3) HealthCare Dashboard

The final dashboard focuses on the health and physical side: average exercise hours, average sleep hours, back/neck pain by gaming platform, physical pain levels, and finally the relationship between daily gaming hours and work productivity, shown through a Scatter Plot.

Code Example: *KPI cards and charts for the HealthCare Dashboard*

```
# Compute interactive health KPI cards based on filters

val_exercise = f"{df_filtered['Exercise_Hours'].mean():.1f}"

val_sleep = f"{df_filtered['Sleep_Hours'].mean():.0f}"

val_gaming_hc = f"{df_filtered['Daily_Gaming_Hours'].mean():.4f}"

col1, col2, col3 = st.columns(3)

with col1:

st.markdown(f'<div class="kpi-card"> AVG of Exercise: | {val_exercise}</div>',
unsafe_allow_html=True)

with col2:

st.markdown(f'<div class="kpi-card"> AVG Sleep hrs: {val_sleep}</div>',
unsafe_allow_html=True)

with col3:

st.markdown(f'<div class="kpi-card"> Daily Gaming Hours: {val_gaming_hc}</div>',
unsafe_allow_html=True)

# Relationship between daily gaming hours and work productivity

fig = px.scatter(df_filtered, x='Work_Productivity', y='Daily_Gaming_Hours')

fig.update_layout(title="Daily Gaming Hours By Work Productivity Score")

st.plotly_chart(fig, use_container_width=True)
```

15. BI Tool Dashboards (Excel, Power BI & Tableau)

In addition to the interactive Streamlit application described in Chapter 12, the same three dashboards — Player, Addiction, and HealthCare — were also implemented using industry-standard business-intelligence tools: Microsoft Excel, Microsoft Power BI, and Tableau. This was done to validate that the analytical findings hold consistently regardless of the tool used, and to demonstrate the same insights through platforms that are more widely used in professional and non-technical settings.

15.1 Excel Dashboards

As part of validating the analytical findings across multiple platforms, the same gaming and mental-health dataset was rebuilt entirely inside Microsoft Excel. Rather than relying on Power BI's native modeling engine, this implementation depends on Power Query (Get & Transform) for data preparation, PivotTables and PivotCharts for aggregation, and Slicers for interactivity. The purpose of reproducing the analysis in Excel is to confirm that the results remain consistent regardless of the tool used, while also demonstrating that the same level of insight can be reached with a tool that is more widely available and familiar to non-technical stakeholders.

Three dashboards were reconstructed using this approach — the Player Dashboard, the Addiction Dashboard, and the HealthCare Dashboard — mirroring the structure adopted in the Power BI implementation. Each dashboard is driven by a shared data model loaded through Power Query, and each relies on synchronized slicers so that filtering one chart updates every other visual on the same sheet.

15.1.1 Data Preparation with Power Query

Before any PivotTable or chart could be built, several raw fields in the dataset needed to be converted into meaningful categorical labels. This transformation work was carried out in Power Query using conditional custom-column formulas, which allowed the categorization logic to be applied consistently to every record before the data was loaded into the workbook's data model. Four calculated columns were created for this purpose.

15.1.1.1 Educational_State

This column classifies each respondent as a Student, a Worker, or a Working_Student by inspecting whether the grades_gpa and work_productivity_score fields contain values. A record with a GPA but no productivity score is labeled Student, a record with a productivity score but no GPA is labeled Worker, and a record containing both is treated as a Working_Student. This derived field later becomes the basis of the Educational_State slicer used across the dashboards.

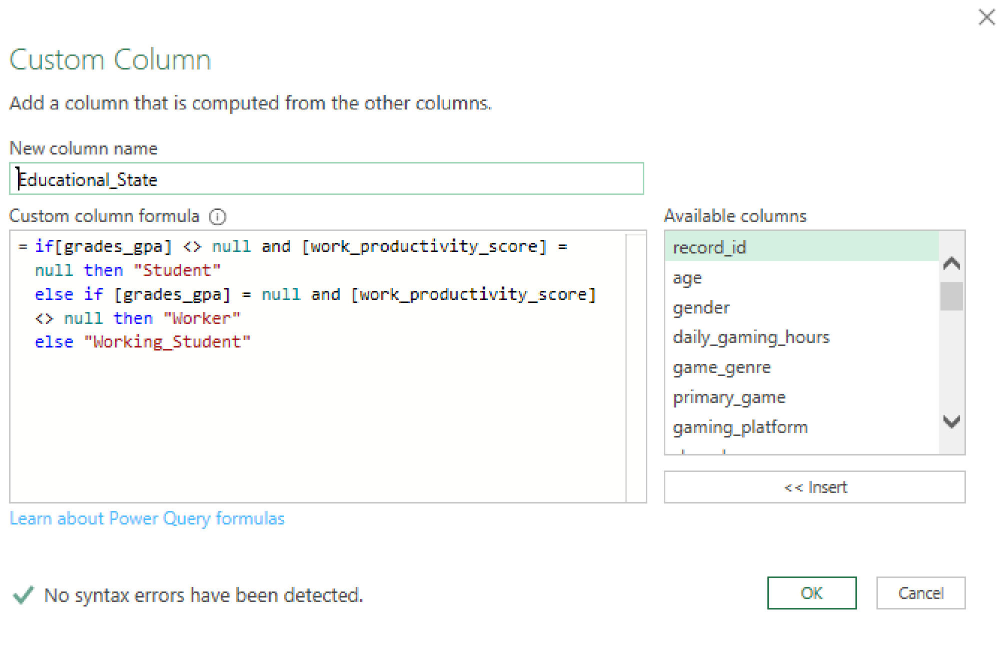


Figure 9.1: Power Query formula for the Educational_State custom column

15.1.1.2 Physical_Pain

To summarize physical discomfort into a single readable indicator, the eye_strain and back_neck_pain boolean fields were combined into one risk category. When both symptoms are present the record is flagged as High_Risk; when only one of the two symptoms is present it is flagged as Moderate; and when neither symptom is reported the record is flagged as No_Risk. This consolidated field is used extensively in the HealthCare Dashboard and also appears as a filtering option there.

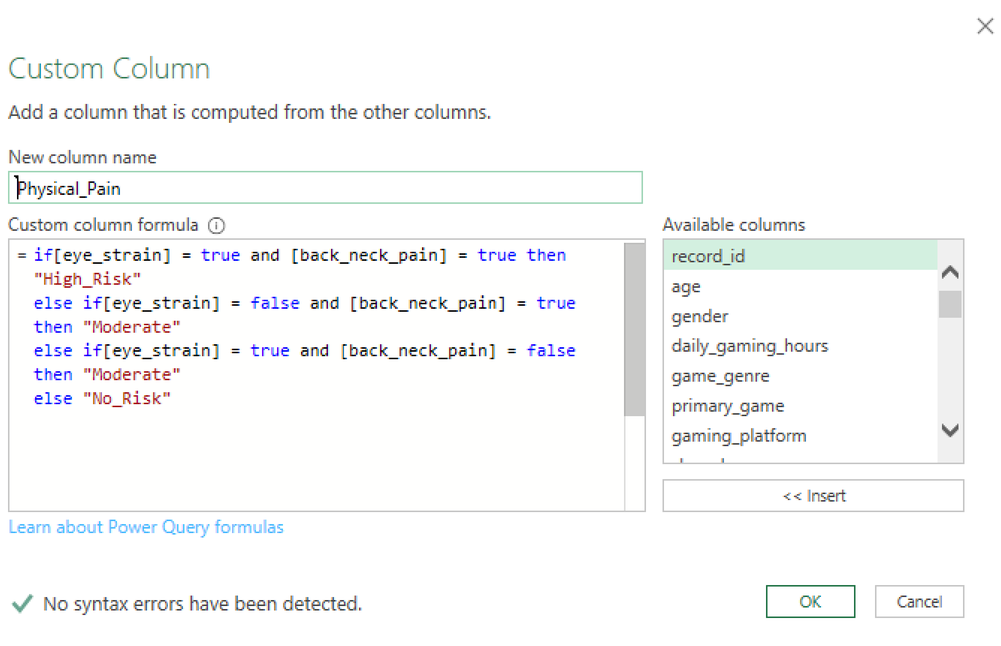


Figure 9.2: Power Query formula for the Physical_Pain custom column

15.1.1.3 Spend_Category

Player spending was grouped into four bands — Low, Mid, High, and Very High — using a statistical threshold rather than fixed cut-off values. The formula compares each player's Total_spent value against the dataset average and its standard deviation: values at or below the average fall into Low, values within one standard deviation above the average fall into Mid, values within two standard deviations fall into High, and anything beyond that is classified as Very High. Basing the thresholds on the distribution of the data, rather than arbitrary numbers, keeps the categorization meaningful even if the dataset changes.

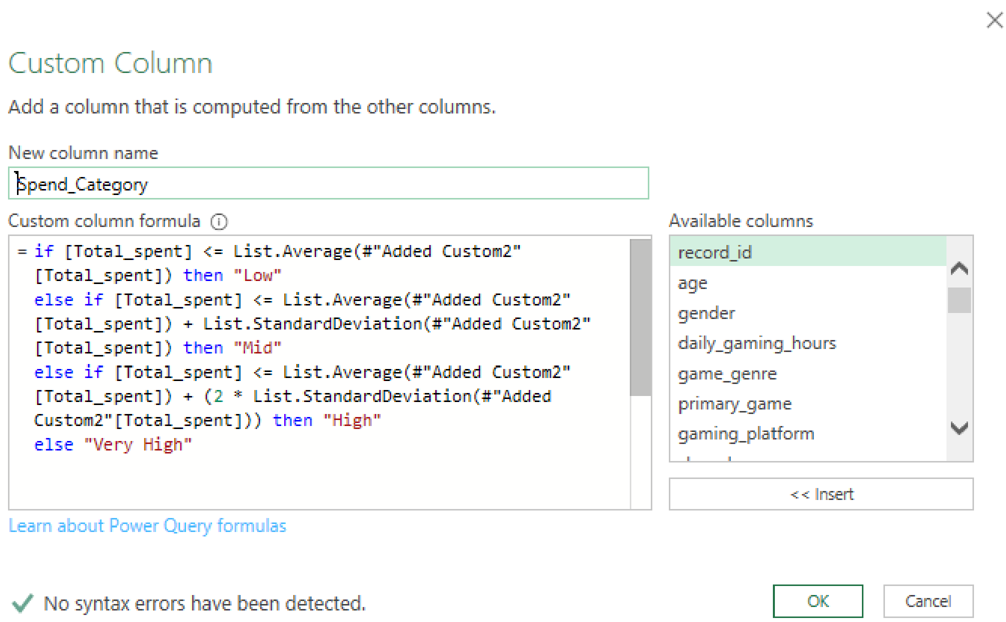


Figure 9.3: Power Query formula for the Spend_Category custom column

15.1.1.4 Gaming_Hours_Category

A comparable statistical approach was applied to daily_gaming_hours to produce a Gaming_Hours_Category field with the same four bands — Low, Mid, High, and Very High — again calculated relative to the average and standard deviation of gaming hours across the dataset. This field makes it possible to compare engagement levels between players using a consistent, data-driven scale instead of raw hour counts.

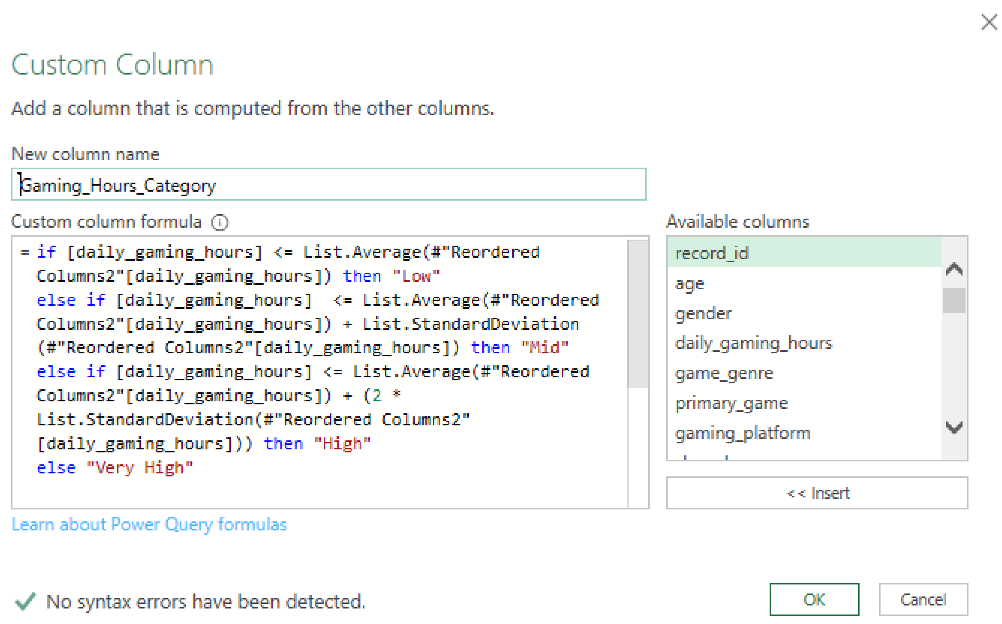


Figure 9.4: Power Query formula for the Gaming_Hours_Category custom column

Once these four columns were validated and loaded, the enriched table was connected to PivotTables, and PivotCharts were built on top of those PivotTables. Slicers were then inserted and linked to multiple pivot tables simultaneously through the Report Connections option, which is what allows a single slicer selection to refresh every chart on a given dashboard sheet at once.

15.1.2 Data Model (Star Schema)

With the calculated columns in place, the workbook's tables were organized into a proper Star Schema rather than being left as one flat sheet. A single central fact table, fact_gaming_mintal_health, holds every measurable event-level field — daily_gaming_hours, sleep_hours, work_productivity_score, mood_state, social_isolation_score, Total_spent, and the four categorical columns produced in Power Query, among others. Six dimension tables surround this fact table, each holding descriptive attributes about one aspect of a record: DIm_Player, Dim_Sleep, Dim_PhysicalStatus, Dim_Game, DIm_Platform, and Dim_Addiction.

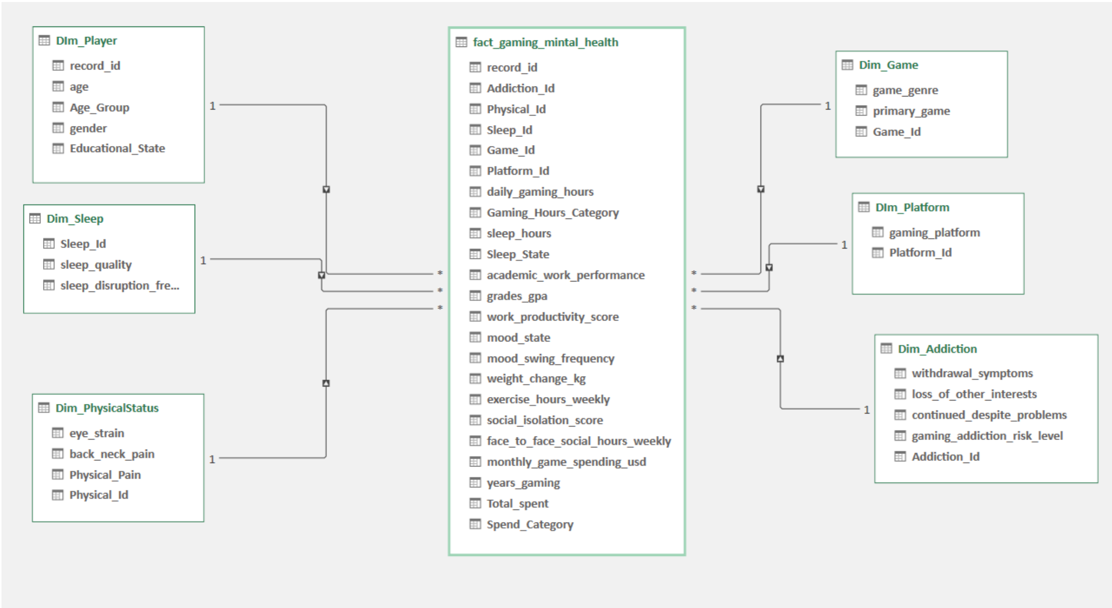


Figure 9.5: Star Schema data model connecting the fact table to six dimension tables

15.1.2.1 Why a Star Schema

Structuring the model this way, instead of keeping one wide table, keeps each dimension's attributes stored once and referenced by key, which avoids repeating the same player, game, or platform details across thousands of rows. It also means that filtering from any dimension — say, selecting a specific Age_Group inside DIm_Player — propagates automatically to the fact table and, from there, to every measure built on top of it. This is the same mechanism that lets a single slicer refresh every chart on a dashboard sheet at once.

15.1.2.2 Relationships

Every relationship in the model runs one-to-many from a dimension table (the '1' side) into the fact table (the '*' side), which is the standard, most performant pattern for this kind of model:

- DIm_Player (1) → fact_gaming_mintal_health (*), joined on record_id — carries age, Age_Group, gender, and Educational_State.
- Dim_Sleep (1) → fact_gaming_mintal_health (*), joined on Sleep_Id — carries sleep_quality and sleep_disruption_frequency.
- Dim_PhysicalStatus (1) → fact_gaming_mintal_health (*), joined on Physical_Id — carries eye_strain, back_neck_pain, and the Physical_Pain category.
- Dim_Game (1) → fact_gaming_mintal_health (*), joined on Game_Id — carries game_genre and primary_game.
- DIm_Platform (1) → fact_gaming_mintal_health (*), joined on Platform_Id — carries gaming_platform.

- Dim_Addiction (1) → fact_gaming_mintal_health (*), joined on Addiction_Id — carries withdrawal_symptoms, loss_of_other_interests, continued_despite_problems, and gaming_addiction_risk_level.

15.1.2.3 Placement of the Engineered Columns

It is worth noting where the four Power-Query columns built earlier ended up in this model. Educational_State sits inside DIm_Player because it describes a lasting trait of the player, not something that changes per session. Physical_Pain sits inside Dim_PhysicalStatus for the same reason — it summarizes a player's condition rather than a single event. Spend_Category and Gaming_Hours_Category, on the other hand, stay inside the fact table itself, since spending and daily gaming hours are values that can differ from one record to the next rather than being fixed traits of the player.

15.1.2.4 What This Model Enables

- A single fact table keeps every measure — spending, gaming hours, productivity, mood, sleep — aligned to the same grain, so numbers from different charts stay comparable.
- Six independent dimensions mean a slicer built on any one of them (platform, age group, addiction risk, sleep quality, and so on) filters the entire fact table without extra formulas.
- Because dimension attributes are stored once per player, game, or platform instead of being repeated on every row, the model stays lighter and faster to refresh than a single flat table would be.
- The same star schema is what lets the Player, Addiction, and HealthCare dashboards share consistent totals — they all pull from the same fact table, just sliced along different dimensions.

15.1.3 Player Dashboard

Figure 9.6: Player Dashboard (Excel)

15.1.3.1 Purpose of the Sheet

The Player Dashboard acts as the entry point of the workbook and gives a broad snapshot of who the players are, how they spend their time, and how much they spend financially. Four slicers — Educational_State, Age_Group, gender, and gaming_platform — sit on the right-hand side of the sheet and are linked to every chart on the page, so an analyst can immediately narrow the view down to, for example, teenage mobile players or working adults on console.

15.1.3.2 KPI Cards

Total Players

Displays a live count of 1.0k records currently included after slicer selections, giving the viewer an immediate sense of how large the underlying sample is.

Total Spent

Sums the Total_spent field across all filtered players, reporting a combined figure of 7.2m and offering a quick read on the aggregate financial footprint of the selected group.

AVG GPA

Averages the grades_gpa field, currently showing 3, and works alongside the Educational_State slicer to check whether academic performance shifts across different player segments.

Daily Gaming Hours

Totals daily_gaming_hours for the filtered records (6.2k), functioning as a companion figure to the spending and academic KPIs so that engagement, cost, and performance can be read side by side.

15.1.3.3 Total Gaming Hours by Game

A horizontal bar chart ranks every title in the dataset — from Mobile Legends through Elden Ring — by cumulative hours played, sorted from lowest to highest. Because the full list of ten games is shown rather than a shortlist, the chart doubles as a reference for which titles dominate playtime (Elden Ring, StarCraft II, and Dota 2 lead the list) and which attract comparatively lighter engagement.

15.1.3.4 Gaming Time by Gender

A donut chart splits total gaming time across Male, Female, and Other categories. The three segments sit close together — Male at 33%, Female at 32%, and Other at 35% — showing that gaming time in this dataset is spread almost evenly across gender groups rather than being concentrated in one segment.

15.1.3.5 Gaming Time by Age Group

A second donut chart breaks the same gaming-time measure down by Teenager, Young_adult, and Adult. The three groups are again close in proportion — Teenager 34%, Young_adult 35%, and Adult 31% — indicating that, unlike some single-platform studies, no single age bracket in this workbook overwhelmingly dominates total playtime.

15.1.3.6 Total Spent by Game Genre

A bar chart compares cumulative spending across genres such as Strategy, RPG, Mobile Games, MOBA, MMO, FPS, and Battle Royale. Strategy tops the ranking at roughly 1.1m, closely followed by MOBA and MMO, while Mobile Games and Battle Royale sit at the lower end. This view highlights which genres draw the heaviest in-game financial commitment rather than simply the most players.

15.1.3.7 Daily Gaming Hours vs Work Productivity Score

A scatter plot plots each player's daily gaming hours against their work productivity score. The point cloud rises through the middle of the range and tapers off at both extremes, suggesting that productivity is not simply lowest among the heaviest gamers or highest among the lightest — the relationship looks more like a mid-range peak than a straight downward trend, which is a useful counterpoint to a purely linear assumption about screen time and output.

15.1.3.8 What This Sheet Adds to the Analysis

- Gives a single-glance summary of sample size, spending, academic standing, and playtime before diving into more specialized sheets.
- Ranks games and genres by two different measures — hours played and money spent — which do not always agree with one another.
- Confirms that gender and age are relatively balanced within this sample, so later findings are not simply an artifact of one dominant group.
- Surfaces a non-linear pattern between gaming hours and productivity that invites closer investigation in later sheets.
- Lets any of the four slicers be combined, so a specific demographic slice can be isolated in seconds.

15.1.3.9 Notable Observations

- Gender and age-group splits are close to even, unlike datasets where one segment dominates.
- Elden Ring, StarCraft II, and Dota 2 accumulate the most total hours among the ten tracked titles.
- Strategy, MOBA, and MMO genres generate the highest cumulative spending.
- Productivity does not fall in a straight line as gaming hours rise — the pattern peaks around the middle of the hour range.

15.1.4 Addiction Dashboard

Figure 9.7: Addiction Dashboard (Excel)

15.1.4.1 Purpose of the Sheet

Where the Player Dashboard paints a general picture, this sheet zeroes in on gaming addiction risk and how it ripples into social life, mood, and spending. Four slicers control the page — gaming_addiction (risk level), Age_Group, gender, and mood_swing_freq — allowing the analyst to isolate, say, severe-risk teenagers or players who report frequent mood swings.

15.1.4.2 KPI Cards

F2F Social Hours

Reports the average number of face-to-face social hours per week for the filtered players, currently 7.7, serving as a baseline for real-world social engagement outside of gaming.

AVG of Social Score

Averages a composite social-isolation score (3.9) that reflects how connected or disconnected players feel, independent of the raw hours they log with others.

AVG Work Productivity

Shows the average work productivity score for the current selection (5.4), letting the analyst check this figure against different addiction-risk slicer settings.

AVG Daily Gaming Hours

Displays average daily gaming hours (6.2) for the filtered group, providing a quick engagement benchmark to compare against the social and productivity KPIs above it.

15.1.4.3 Gaming Addiction Risk by Age Group

A clustered column chart breaks down addiction-risk counts — High, Low, Moderate, and Severe — separately for Adults, Teenagers, and Young_adults. Young_adults show by far the largest counts across all four risk bands, most visibly in the Low category (312), while Adults report the smallest counts overall. This points to young adulthood as the life stage where gaming-related risk behavior is most concentrated within this sample.

15.1.4.4 How Addiction Risk Affects Social Life

A grouped bar chart compares two metrics — average social isolation score and average face-to-face social hours per week — across the four addiction-risk categories. The Low-risk group posts the highest face-to-face hours and one of the lower isolation scores, while the High-risk group shows a markedly lower isolation score paired with reduced face-to-face time, a combination worth flagging even though it runs counter to a simple 'more risk equals more isolation' assumption and merits a closer statistical look.

15.1.4.5 Total Spending by Mood State

A line chart tracks cumulative spending across nine mood states, ordered from Excited (3.1k) through Anxious (10.2k). Spending climbs steadily as the mood states move toward more negative or agitated emotional labels, with Irritable and Anxious states associated with the highest cumulative totals — a pattern consistent with the idea that spending can rise alongside emotional discomfort rather than only during positive or celebratory moods.

15.1.4.6 How Mood Affects Gaming Time

A horizontal bar chart lists average daily gaming hours for each mood label. Most negative or agitated states — Anxious, Irritable, Restless, Depressed, Withdrawn, and Angry — cluster tightly around 7 hours, Euphoric sits at 6, while Normal and Excited drop to 4 hours each. In other words, calmer or more positive moods are associated with noticeably shorter gaming sessions than tense or low moods in this dataset.

15.1.4.7 Slicer-Driven Exploration

Because all four charts share the same set of slicers, the sheet supports rapid comparison work — for instance, switching the gaming_addiction slicer to Severe immediately recalculates every visual, from the age-group breakdown to the mood-spending line, to reflect only that subgroup, without touching a single formula.

15.1.4.8 What This Sheet Adds to the Analysis

- Breaks addiction risk down by life stage, pinpointing young adulthood as the most affected group.
- Pairs a behavioral metric (face-to-face hours) with a perceptual one (isolation score) to give a fuller social picture than either alone.
- Links spending directly to emotional state rather than only to demographics or genre.
- Shows that gaming duration itself tracks mood, not just money spent.

- Keeps every chart synchronized to the same slicer panel for fast, consistent filtering.

15.1.4.9 Notable Observations

- Young_adults dominate every addiction-risk band, especially the Low-risk count.
- Face-to-face social hours and isolation scores do not move in perfect lockstep across risk categories.
- Spending rises as mood shifts from Excited toward Anxious, peaking in the more negative emotional states.
- Negative or restless moods correspond to roughly 7 hours of daily play, versus 4 hours for calmer states.

15.1.5 HealthCare Dashboard

Figure 9.8: HealthCare Dashboard (Excel)

15.1.5.1 Purpose of the Sheet

This final sheet turns attention to the bodily side effects of gaming — exercise habits, sleep, and physical pain — rather than the social or financial angles covered elsewhere. Four slicers, Age_Group, gender, gaming_platform, and Physical_Pain, control the page, making it possible to compare, for instance, console players flagged as High_Risk against mobile players flagged as No_Risk.

15.1.5.2 KPI Cards

AVG of Exercise

Averages weekly exercise hours for the filtered players (6.9), acting as the baseline physical-activity figure for the sheet.

AVG Sleep hrs

Reports average nightly sleep hours (6) for the current selection, a figure that can be cross-checked against the sleep-quality chart further down the sheet.

Daily Gaming Hours

Shows the average daily gaming hours (6.1514) for the filtered group, again serving as the common engagement benchmark against which the exercise and sleep KPIs are read.

15.1.5.3 Back/Neck Pain by Gaming Platform

A clustered bar chart compares reported back or neck pain (TRUE vs FALSE) across Console, Mobile, Multi-platform, and PC. Every platform shows more FALSE than TRUE responses, but the gap narrows noticeably on PC (164 vs 77), which reports proportionally the highest share of pain complaints among the four platforms, while Console shows the widest gap between the two categories.

15.1.5.4 Physical Pain Risk by Daily Gaming Hours

A donut chart summarizes the Physical_Pain field created earlier in Power Query across the filtered players: High_Risk accounts for 43%, Moderate for 37%, and No_Risk for only 20%. With well over half of the sample landing in the Moderate-or-High bands, physical discomfort appears to be the norm rather than the exception among the players represented here.

15.1.5.5 Mood State by Exercise Hours

A horizontal bar chart totals exercise hours contributed by players in each mood state. Normal contributes by far the largest share (1371.80 hours), followed by Irritable (979.00) and Anxious (1069.70), while Withdrawn (271.60) and Excited (321.90) contribute the least — largely reflecting how many players fall into each mood category as much as how active any single player is.

15.1.5.6 Sleep Quality by Sleep Hours

A bar chart compares average sleep hours across five sleep-quality labels — Fair, Good, Insomnia, Poor, and Very Poor. Good sleep quality corresponds to the highest average duration (6.85 hours), Fair sits close behind (6.04), while Insomnia (4.75) and Very Poor (4.88) post the shortest averages, confirming that the sleep-quality labels line up sensibly with actual reported sleep duration.

15.1.5.7 Daily Gaming Hours vs Work Productivity Score

A second scatter plot, matching the one on the Player Dashboard but recalculated for this sheet's slicer selections, again shows productivity building up through the middle of the gaming-hours range before tapering at the higher end — reinforcing, from a health-focused lens, that heavier gaming does not translate into a simple straight-line productivity decline in this dataset.

15.1.5.8 Slicer-Driven Exploration

As with the other two sheets, every chart here responds to the same set of slicers at once. Setting the Physical_Pain slicer to High_Risk, for example, instantly recalculates the exercise, sleep, and platform charts to describe only that at-risk subgroup, which is useful for spotting whether pain-affected players also sleep worse or exercise less.

15.1.5.9 What This Sheet Adds to the Analysis

- Connects the gaming platform used to the likelihood of reporting back or neck pain.
- Condenses two separate pain symptoms into one composite risk read via the Physical_Pain donut.
- Validates that self-reported sleep-quality labels correspond to sensible differences in actual sleep duration.
- Cross-references mood, exercise, and gaming hours on one sheet rather than treating them separately.
- Repeats the productivity-vs-hours scatter under a health lens, adding weight to the pattern seen on the Player Dashboard.

15.1.5.10 Notable Observations

- PC players report the narrowest gap between pain and no-pain responses among the four platforms.
- Only one in five filtered players falls into the No_Risk physical-pain band.
- Good sleep quality lines up with roughly 6.85 average sleep hours, the highest of the five quality labels.
- The gaming-hours-versus-productivity pattern seen here echoes the one found on the Player Dashboard, adding confidence to that finding.

15.1.6 Cross-Dashboard Findings

Reviewing all three Excel sheets together produces a coherent narrative that lines up closely with what the Power BI build showed, despite being generated with a completely different toolset.

Demographics are broad rather than concentrated: gender and age splits on the Player Dashboard stay close to even, and **Young_adults** emerge as the group most exposed to addiction risk on the second sheet, echoing their prominence in overall engagement.

Money follows genre more than platform: Strategy, MOBA, and MMO titles draw the heaviest cumulative spending, a pattern that holds regardless of which gaming platform a player prefers.

Emotional state is tied to both spending and playtime: cumulative spending rises as mood shifts toward Anxious or Irritable, and average daily gaming hours are noticeably longer during negative or restless mood states than during calmer ones.

Physical strain is common rather than rare: roughly four out of five filtered players carry at least a Moderate physical-pain classification, and pain complaints are proportionally most common among PC players.

Productivity does not decline in a straight line with gaming hours: both scatter charts — on the Player and HealthCare sheets — show productivity climbing through the middle of the hour range before falling off, a nuance that a single summary statistic would have missed.

Taken together, these results confirm that the conclusions reached in the Power BI dashboards are not an artifact of that specific tool. Rebuilding the same logic with Power Query and PivotCharts in Excel reproduces the same demographic, financial, emotional, and physical patterns, which strengthens confidence in the underlying findings.

15.1.7 Conclusion

This Excel implementation demonstrates that a spreadsheet-native workflow — Power Query for feature engineering, PivotTables for aggregation, PivotCharts for visualization, and linked Slicers for interactivity — can reproduce the analytical depth of a dedicated business-intelligence tool without requiring specialized software. The three sheets built here, the Player Dashboard, the Addiction Dashboard, and the HealthCare Dashboard, cover the same demographic, behavioral, and health-related territory explored in the Power BI version, and they arrive at consistent conclusions.

Beyond validating the results across tools, this version of the workbook has a practical advantage: because it relies only on Microsoft Excel, it can be opened, filtered, and explored by stakeholders who may not have access to Power BI, making the findings easier to circulate and easier for a non-technical audience to interact with directly.

15.2 Power BI – Player Dashboard

Figure 13.1: Power BI – Player Dashboard

15.2.1 Overview

The Player Dashboard provides a comprehensive overview of the gaming community. It focuses on demographic characteristics, gaming behavior, spending patterns, and productivity indicators. The dashboard enables users to understand the overall structure of the dataset before exploring more specialized analyses.

Interactive slicers allow filtering the data by gender, age group, gaming platform, addiction level, and other attributes. Consequently, every visualization updates dynamically according to the selected filters, making the dashboard highly interactive and suitable for exploratory analysis.

15.2.2 KPI Cards

The first section of the dashboard contains four Key Performance Indicators (KPIs). These cards provide a quick summary of the most important numerical metrics.

Total Players

This KPI represents the total number of players included in the dataset.

It provides an overall understanding of the dataset size and serves as a reference for interpreting all subsequent charts.

A large sample size increases the reliability of the analytical findings and ensures that the visualizations represent a broad gaming population.

Total Spending

This KPI displays the total lifetime spending of all players.

The value is calculated using the Total Spent feature that was created during the preprocessing stage.

This indicator helps identify the overall economic impact of gaming within the dataset and supports financial behavior analysis.

Total Gaming Hours

This card shows the accumulated gaming hours for all players.

It provides an estimate of overall gaming engagement and helps compare different player groups after applying dashboard filters.

Average Work Productivity Score

This KPI represents the average work productivity score across all players.

It allows decision-makers to investigate whether excessive gaming may influence workplace performance when compared with other variables.

15.2.3 Gender Distribution

The donut chart illustrates the percentage distribution of players based on gender.

Using a donut chart allows users to compare category proportions quickly while maintaining a clean dashboard layout.

The visualization demonstrates whether gaming participation is balanced across genders or dominated by a particular group.

This analysis is particularly useful when investigating whether behavioral patterns differ between male and female players.

15.2.4 Top Five Games by Average Gaming Hours

The horizontal bar chart ranks the five games with the highest average daily gaming hours.

Sorting the chart in descending order allows users to immediately identify the games that require the greatest time commitment.

This visualization helps understand player preferences and identifies games that may contribute to prolonged gaming sessions.

15.2.5 Age Group Distribution

Players were divided into three age categories:

- Teenager
- Young Adult
- Adult

The bar chart compares the number of players within each category.

The analysis shows that **Young Adults** represent the largest portion of the gaming community.

This finding indicates that gaming activity is most common among individuals between approximately 19 and 28 years old.

Understanding the dominant age group helps organizations target awareness campaigns and health recommendations more effectively.

15.2.6 Total Spending by Game Genre

This visualization compares lifetime spending across different game genres.

Instead of focusing on the number of players, this chart measures the financial contribution of each genre.

Genres such as MMO and Strategy generate considerably higher spending than others, suggesting that these games encourage long-term engagement and in-game purchases.

Such information is valuable for both business analysis and behavioral studies.

15.2.7 Gaming Hours vs Work Productivity

The scatter plot investigates the relationship between daily gaming hours and work productivity.

Each point represents one player.

Scatter plots are particularly useful because they reveal trends, clusters, and possible correlations that may not be visible in summary statistics.

If gaming hours increase while productivity decreases, a negative relationship may exist.

Conversely, if no clear pattern appears, gaming hours alone may not explain productivity differences.

This visualization therefore supports exploratory analysis rather than direct causal conclusions.

15.2.8 Dashboard Benefits

The Player Dashboard provides several important advantages:

- Offers a comprehensive overview of the gaming community.
- Simplifies demographic analysis.
- Identifies the most popular games.
- Highlights spending behavior across genres.
- Supports productivity analysis.
- Enables interactive filtering for detailed exploration.

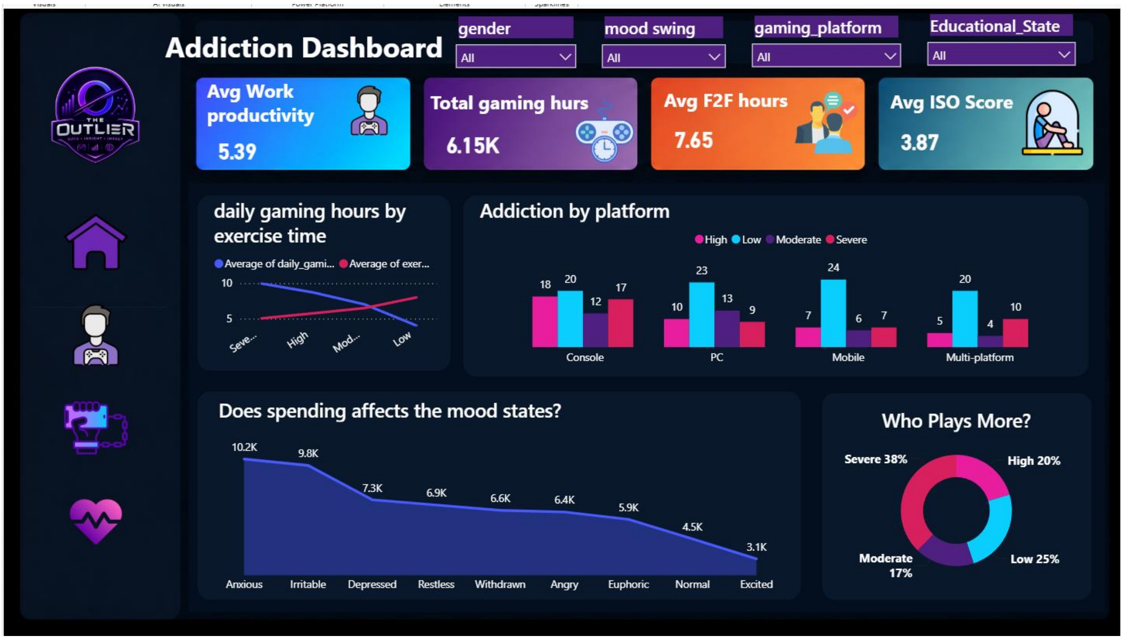
15.2.9 Key Insights

Several important findings can be observed from this dashboard:

- Young Adults represent the largest player group.
- Certain game genres generate significantly higher spending.
- Gaming engagement varies considerably across players.
- Work productivity differs among players with different gaming habits.
- Interactive filtering allows deeper investigation into specific player segments.

15.3 Power BI – Addiction Dashboard

Figure 13.2: Power BI – Addiction Dashboard



15.3.1 Overview

The Addiction Dashboard focuses on analyzing gaming addiction and its relationship with players' daily activities, social interaction, mood, and spending behavior. Unlike the Player Dashboard, which provides a general overview of the dataset, this dashboard specifically investigates how different levels of gaming addiction influence various aspects of players' lives.

Interactive filters enable users to explore addiction patterns by age group, gender, gaming platform, and other demographic variables. As a result, all visualizations update dynamically, allowing users to compare different groups and identify behavioral trends.

The primary objective of this dashboard is to determine whether increasing gaming addiction is associated with changes in productivity, physical activity, emotional well-being, and financial spending.

15.3.2 KPI Cards

The dashboard begins with four KPI cards that summarize the most important addiction-related metrics.

Average Work Productivity Score

This KPI displays the average productivity score of all selected players.

Monitoring this value helps evaluate whether gaming addiction is associated with reduced workplace or academic performance.

By comparing this KPI across different addiction levels, users can observe whether productivity changes as gaming intensity increases.

Total Gaming Hours

This KPI represents the total gaming hours accumulated by all filtered players.

It provides a direct measure of gaming engagement and allows comparisons between addiction categories.

Players with higher addiction levels generally contribute a larger proportion of the total gaming hours.

Average Face-to-Face Social Hours

This KPI measures the average number of hours players spend interacting with others in real-life social activities.

It serves as an indicator of social engagement outside the gaming environment.

Lower values may suggest that increased gaming reduces face-to-face interaction.

Average Social Isolation Score

The Social Isolation Score summarizes how socially isolated players feel.

Unlike social hours, which measure actual interaction time, this metric reflects the player's perceived level of isolation.

Higher scores indicate stronger feelings of loneliness or reduced social connectedness.

15.3.3 Gaming Hours vs Exercise Hours

This visualization compares average gaming hours with weekly exercise hours.

The chart helps determine whether physically active players spend less time gaming.

Exercise is considered an important health indicator, making this comparison valuable for understanding lifestyle balance.

If exercise hours decrease while gaming hours increase, it may suggest that excessive gaming replaces physical activity.

However, the dashboard should be interpreted as identifying relationships rather than proving direct causation.

15.3.4 Gaming Addiction by Platform

Different gaming platforms attract different player behaviors.

This visualization compares addiction levels across platforms such as:

- PC
- Console
- Mobile
- Multi-platform

The chart allows users to identify whether particular platforms are associated with higher addiction rates.

For example, players who frequently switch between multiple platforms may accumulate more gaming hours than those using only one platform.

Such findings may assist researchers in identifying platform-specific behavioral patterns.

15.3.5 Total Spending by Mood State

Mood plays an important role in gaming behavior.

This visualization compares players' total spending across different emotional states.

Possible mood categories include:

- Happy
- Neutral
- Stressed
- Sad

Analyzing financial behavior alongside mood may reveal whether emotional conditions influence purchasing decisions within games.

For instance, some players experiencing stress may spend more money on gaming as a coping mechanism.

Although this relationship cannot confirm psychological causes, it provides valuable insight for further investigation.

15.3.6 Distribution of Addiction Risk Levels

The dashboard also illustrates the distribution of players across addiction categories.

The categories include:

- Low Risk
- Moderate Risk
- Severe Risk

This visualization provides a quick overview of the overall addiction profile of the gaming community.

Understanding the proportion of players in each category helps prioritize intervention strategies and awareness campaigns.

15.3.7 Interactive Filtering

The Addiction Dashboard supports interactive filtering using several attributes, including:

- Gender
- Age Group
- Gaming Platform
- Addiction Level
- Educational Status

Applying filters enables users to explore specific subsets of players without modifying the underlying dataset.

This interactive capability makes the dashboard useful for both researchers and decision-makers.

15.3.8 Dashboard Benefits

The Addiction Dashboard provides several important advantages:

- Measures gaming addiction through multiple indicators.
- Connects addiction with productivity and social interaction.
- Supports behavioral and psychological analysis.
- Helps identify high-risk player groups.
- Enables dynamic comparison across demographic categories.
- Provides interactive exploration using slicers and filters.

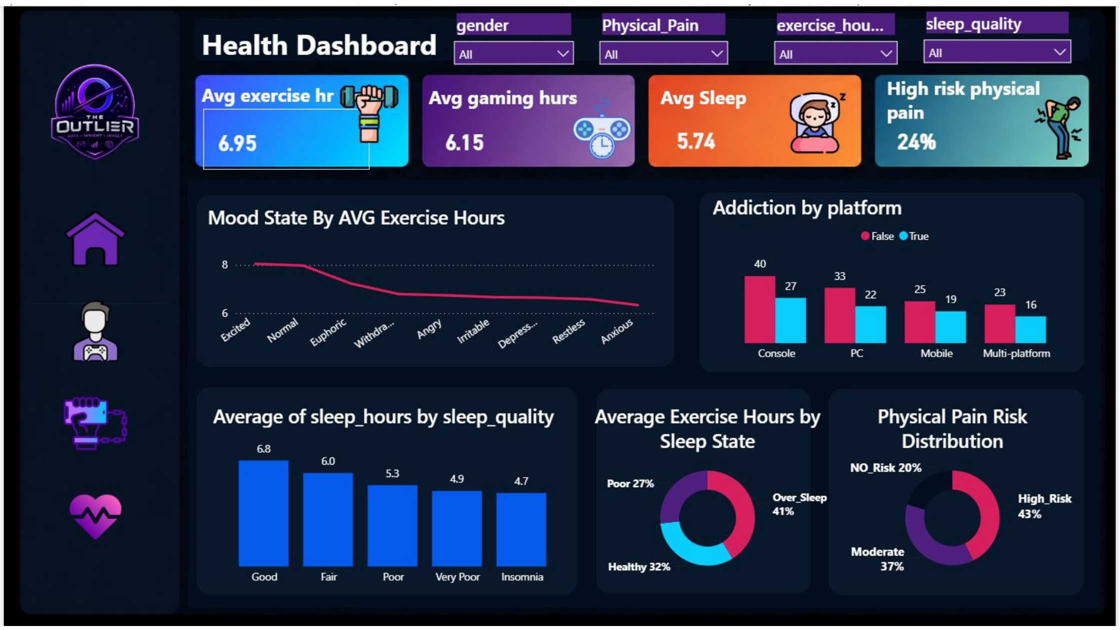
15.3.9 Key Insights

Several important observations can be drawn from this dashboard:

- Players with higher addiction levels generally spend more time gaming.
- Increased gaming engagement is often accompanied by reduced face-to-face social interaction.
- Mood appears to influence spending behavior in certain player groups.
- Exercise levels tend to differ between addiction categories.
- Gaming platform may influence the distribution of addiction risk.
- Interactive analysis allows researchers to investigate addiction patterns efficiently.

15.4 Power BI – Healthcare Dashboard

Figure 13.3: Power BI – Healthcare Dashboard



15.4.1 Overview

The Healthcare Dashboard focuses on the physical and health-related effects associated with gaming behavior. While the previous dashboards emphasized player demographics and addiction patterns, this dashboard investigates how gaming habits influence sleep quality, physical activity, physical pain, and overall well-being.

The dashboard combines multiple health indicators into one interactive environment, enabling users to identify relationships between gaming behavior and physical health outcomes. Interactive slicers allow users to analyze specific player groups, making the dashboard suitable for both research and decision-making purposes.

The main objective of this dashboard is to determine whether excessive gaming is associated with changes in exercise habits, sleep quality, and physical discomfort.

15.4.2 KPI Cards

The Healthcare Dashboard begins with four KPI cards summarizing the most important health-related indicators.

Average Exercise Hours

This KPI represents the average number of exercise hours performed by players each week.

Regular exercise is considered an important indicator of a healthy lifestyle. Monitoring this value helps determine whether highly active gamers maintain sufficient physical activity.

A lower average exercise time may indicate that gaming replaces time normally dedicated to physical activities.

Average Gaming Hours

This KPI displays the average daily gaming hours for all selected players.

It serves as the primary indicator of gaming engagement and provides a useful reference when comparing other health variables such as sleep and physical pain.

Average Sleep Hours

Sleep is one of the most important health indicators included in the dataset.

This KPI calculates the average number of hours players sleep each day.

Maintaining healthy sleep duration is essential for both physical recovery and mental well-being. Comparing this metric with gaming hours helps identify possible behavioral patterns.

High Physical Pain Risk

This KPI represents the percentage of players classified as **High Risk** according to the Physical Pain feature created during preprocessing.

The classification combines eye strain and back/neck pain into one health indicator.

Higher values suggest that physical discomfort is common among players with intensive gaming habits.

15.4.3 Average Exercise Hours by Mood State

This visualization compares the average weekly exercise hours across different mood states.

Mood categories may include:

- Happy
- Neutral
- Stressed
- Sad

The objective is to explore whether emotional well-being influences physical activity.

Players reporting positive moods often demonstrate healthier exercise habits than those experiencing stress or negative emotions.

This comparison supports a broader understanding of lifestyle behaviors beyond gaming alone.

15.4.4 Gaming Addiction by Platform

This chart compares addiction cases across gaming platforms.

Different platforms encourage different gaming behaviors due to accessibility, game design, and playing habits.

Comparing addiction distribution across platforms helps identify whether certain gaming environments are associated with increased addiction risk.

These findings may support future studies investigating platform-specific behavioral patterns.

15.4.5 Average Sleep Hours by Sleep Quality

Sleep quality is closely associated with gaming behavior.

This visualization compares the average number of sleep hours for different sleep quality categories.

The chart allows users to verify whether players reporting poor sleep quality also experience shorter sleeping durations.

Such information helps validate the sleep classifications generated during the preprocessing stage.

15.4.6 Sleep State Distribution

The donut chart illustrates the proportion of players classified into the following sleep categories:

- Poor Sleep
- Healthy Sleep
- Over Sleep

Using a donut chart allows users to compare category proportions quickly.

The visualization provides an overall understanding of sleeping habits within the gaming community.

If the majority of players belong to the Healthy category, it suggests relatively balanced sleeping patterns, whereas a larger Poor Sleep category may indicate potential health concerns.

15.4.7 Physical Pain Risk Distribution

The Physical Pain visualization summarizes the percentage of players within each physical risk category.

The categories include:

- High Risk
- Moderate Risk
- No Risk

This chart combines multiple health symptoms into one clear indicator, making interpretation significantly easier.

Researchers and healthcare professionals can quickly identify how widespread physical discomfort is among gamers.

15.4.8 Interactive Features

Like the previous dashboards, the Healthcare Dashboard supports interactive filtering using several attributes such as:

- Gender
- Age Group
- Gaming Platform
- Sleep State
- Addiction Risk Level
- Educational Status

All charts and KPI cards update instantly according to the selected filters, allowing users to perform customized analyses without modifying the underlying dataset.

15.4.9 Dashboard Benefits

The Healthcare Dashboard offers several important benefits:

- Monitors players' physical well-being.
- Evaluates sleeping behavior.
- Analyzes exercise habits.
- Measures physical pain risk.
- Supports health-related decision-making.
- Enables interactive exploration of health indicators.

15.4.10 Key Insights

Several important observations can be obtained from this dashboard:

- Players with higher gaming activity may experience increased physical discomfort.
- Sleep quality varies across different gaming behaviors.
- Regular exercise appears to be associated with healthier mood states.
- Physical pain remains an important concern among highly engaged players.
- Interactive filtering allows detailed health analysis for specific player groups.

15.5 Overall Power BI Dashboard Findings

After analyzing all three dashboards, several important findings can be summarized.

First, **Young Adults** represent the largest segment of the gaming community, indicating that this age group is the most actively engaged in gaming activities.

Second, spending behavior varies considerably across game genres. Strategy and MMO games tend to generate higher lifetime spending, suggesting that players invest more financially in long-term and competitive gaming experiences.

Third, gaming addiction is associated with noticeable behavioral changes. Players with higher addiction levels generally spend more time gaming, report lower face-to-face social interaction, and display higher social isolation scores.

Fourth, the Healthcare Dashboard highlights the importance of maintaining healthy lifestyle habits. Players with intensive gaming patterns are more likely to experience physical discomfort, reduced exercise time, and changes in sleep behavior.

Finally, the integration of interactive dashboards enables users to transform large volumes of raw data into meaningful visual insights. Decision-makers can quickly explore player characteristics, compare different demographic groups, and identify important trends without manually analyzing thousands of records.

15.6 Power BI Conclusion

The interactive dashboards developed in this project successfully transformed the processed gaming dataset into an intuitive visual analytics platform. By integrating demographic information, behavioral patterns, addiction indicators, and health-related metrics, the dashboards provide a comprehensive understanding of the gaming community.

The use of interactive KPI cards, charts, and slicers allows users to explore the data efficiently and supports evidence-based decision-making. Rather than relying solely on statistical tables, stakeholders can identify trends, compare player groups, and evaluate the impact of gaming habits through dynamic visualizations.

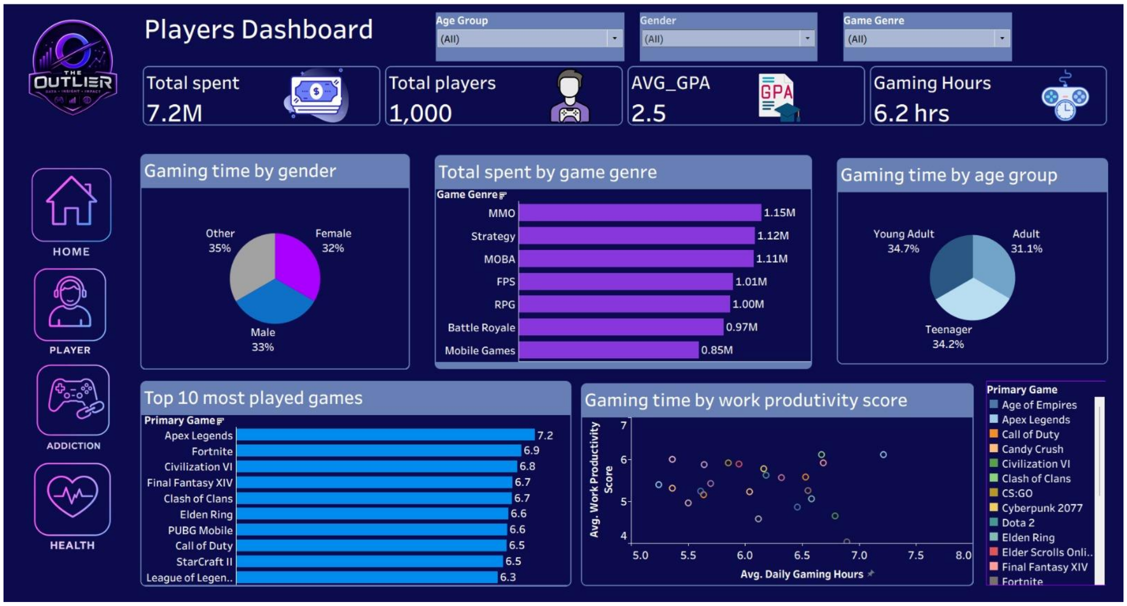
Overall, the dashboards demonstrate how business intelligence tools such as **Power BI** can convert complex datasets into meaningful insights, making them valuable for researchers, healthcare professionals, and decision-makers interested in understanding gaming behavior and its effects on mental and physical well-being.

15.7 Tableau Dashboards

The Tableau dashboards were developed to provide an additional interactive visualization environment for analyzing the gaming and mental health dataset. While the Power BI dashboards emphasize business intelligence reporting, Tableau offers a highly visual and exploratory approach that enables users to investigate relationships between player behavior, gaming addiction, and health indicators through interactive charts and dynamic filters.

15.7.1 Players Dashboard

Figure 13.4: Tableau – Players Dashboard



Purpose

The Players Dashboard provides an overview of the gaming population by examining player demographics, gaming preferences, academic performance, and spending behavior. It enables users to identify the characteristics of different player groups and explore how gaming habits differ according to age, gender, and game genre.

Dashboard Components

KPI Cards

The dashboard summarizes the dataset using four primary performance indicators:

- Total Spending
- Total Players
- Average GPA
- Average Daily Gaming Hours

These KPIs provide a quick overview of the selected data before users explore the detailed visualizations.

Visualizations

Gaming Time by Gender

This pie chart illustrates how gaming time is distributed among male, female, and other players. It helps determine whether gaming participation differs across gender groups.

Total Spending by Game Genre

The horizontal bar chart ranks game genres according to their total spending. It highlights which categories generate the greatest financial contribution from players.

Gaming Time by Age Group

This visualization compares gaming activity among Teenagers, Young Adults, and Adults, allowing users to identify the most active age categories.

Top 10 Most Played Games

The chart ranks games based on their average gaming hours, making it easy to identify the most frequently played titles within the dataset.

Gaming Hours vs Work Productivity

A scatter plot explores the relationship between daily gaming hours and work productivity scores. This visualization helps identify trends or possible correlations between gaming intensity and productivity.

Dashboard Benefits

The Players Dashboard enables users to:

- Examine demographic characteristics of gamers.
- Compare spending across different game genres.
- Identify the most popular games.
- Explore potential relationships between gaming behavior and productivity.
- Perform interactive filtering using Age Group, Gender, and Game Genre.

Key Findings

The dashboard reveals several important observations:

- Young Adults represent the largest proportion of active players.
- MMO and Strategy games account for the highest total spending.
- Apex Legends records the highest average gaming hours among all games.
- Moderate gaming activity shows only a limited relationship with work productivity.

15.7.2 Addiction Dashboard

Figure 13.5: Tableau – Addiction Dashboard

Purpose

The Addiction Dashboard focuses on understanding gaming addiction and its effects on players' daily lives. It combines behavioral, financial, and social indicators to illustrate how addiction levels influence lifestyle patterns and player engagement.

Dashboard Components

KPI Cards

The dashboard summarizes addiction-related information using four key indicators:

- Average Work Productivity
- Face-to-Face Social Hours
- Average Gaming Hours
- Average Social Isolation Score

These metrics provide a high-level understanding of the selected player group before exploring detailed analyses.

Visualizations

Average Gaming Hours vs Exercise Hours

A line chart compares gaming hours with weekly exercise time across addiction risk levels. It illustrates how physical activity changes as gaming addiction increases.

Addiction by Platform

A clustered column chart compares addiction levels across Console, PC, Mobile, and Multi-platform players.

Spending by Mood States

This visualization analyzes players' spending behavior according to different emotional states, highlighting possible links between mood and financial activity.

Addiction by Total Spending

The pie chart summarizes total spending for different addiction categories, allowing users to compare financial commitment across addiction levels.

Dashboard Benefits

The Addiction Dashboard helps users to:

- Compare exercise habits with gaming intensity.
- Investigate addiction across gaming platforms.
- Analyze the relationship between emotions and spending.
- Evaluate social interaction among different addiction groups.
- Explore behavioral patterns using interactive filters.

Key Findings

The dashboard highlights several important trends:

- Gaming hours increase steadily as addiction risk becomes more severe.
- Weekly exercise tends to decrease among highly addicted players.
- Players classified as severe risk contribute the highest overall spending.
- Increased addiction is associated with lower face-to-face social interaction and higher social isolation.

15.7.3 Health Dashboard

Figure 13.6: Tableau – Health Dashboard

Purpose

The Health Dashboard evaluates the impact of gaming behavior on players' physical health and lifestyle. It integrates sleep, exercise, physical pain, and mood indicators to assess the potential health consequences associated with prolonged gaming.

Dashboard Components

KPI Cards

The dashboard presents three main health indicators:

- Average Gaming Hours
- Average Exercise Hours
- Average Sleep Hours

These KPIs provide a concise overview of players' overall health-related behavior.

Visualizations

Average Exercise Hours by Sleep States

This bar chart compares weekly exercise hours among players with different sleep states, helping evaluate lifestyle balance.

Physical Pain by Gaming Hours

The pie chart summarizes average gaming hours across different physical pain categories, showing how gaming intensity relates to physical discomfort.

Mood Effects on Gaming Hours

A line chart illustrates how average gaming hours vary according to players' emotional states.

Gaming Platform by Back and Neck Pain

This clustered column chart compares reported back and neck pain across different gaming platforms.

Sleep Quality by Sleep Hours

The visualization compares average sleeping hours among different sleep quality categories, providing insight into players' sleeping behavior.

Dashboard Benefits

The Health Dashboard allows users to:

- Monitor sleep and exercise habits.
- Analyze physical pain associated with gaming.
- Compare health indicators across gaming platforms.
- Investigate the influence of mood on gaming behavior.
- Identify potential health risks among different player groups.

Key Findings

The dashboard provides several valuable insights:

- Players reporting higher physical pain generally spend more time gaming.
- Good sleep quality is associated with longer sleeping duration.
- Mood has a noticeable influence on daily gaming hours.
- Intensive gaming behavior is frequently accompanied by physical discomfort.
- Health indicators vary across gaming platforms and player groups.

15.7.4 Overall Tableau Findings

The Tableau dashboards provide an interactive analytical environment that complements the Power BI implementation by emphasizing visual exploration and flexible data analysis. Through dynamic filters and intuitive visualizations, users can investigate demographic characteristics, addiction patterns, spending behavior, and health indicators from multiple perspectives.

Overall, the dashboards demonstrate that excessive gaming is associated with increased spending, higher addiction risk, reduced physical activity, and greater health concerns. The interactive nature of Tableau also enables users to discover hidden patterns and compare different player groups efficiently, making it a valuable tool for exploratory data analysis and decision support.

16. Dashboard Analysis & Key Insights Summary

16.1 Introduction

After preparing the dataset, interactive dashboards were developed to visualize gaming behavior and mental health indicators.

16.2 Key Performance Indicators (KPIs)

The dashboards present several important KPIs, including:

- Total Players
- Average Daily Gaming Hours
- Average Sleep Hours
- Average Stress Level
- Average Social Interaction Score
- Addiction Risk Distribution

16.3 Dashboard Analysis

The dashboards analyze several important relationships, including:

- Gaming Hours vs. Stress Level
- Sleep Duration vs. Mental Health
- Gaming Behavior by Age Group
- Addiction Risk by Gender
- Monthly Gaming Spending
- Social Interaction vs. Gaming Habits

Interactive filters allow users to explore the data by age, gender, occupation, and gaming platform.

16.4 Key Insights

The analysis indicates that longer gaming sessions are generally associated with higher stress levels and greater addiction risk. Players with healthier sleep habits tend to report better mental health indicators. Differences in gaming behavior are also observed across age groups and gaming platforms.

16.5 Recommendations

Based on the analysis, the following recommendations are proposed:

- Encourage balanced gaming habits.
- Promote healthy sleep routines.
- Raise awareness about gaming addiction.
- Continue monitoring player behavior using data-driven approaches.

16.6 Chapter Summary

This chapter presented the dashboards, analytical findings, and recommendations derived from the data.

17. Strategic Actionable Recommendations

1. Implement Strict Screen-Time Hard Limits

Gamers should leverage console and PC built-in digital well-being tools to enforce a maximum cap of 3 to 3.5 hours of daily gaming, to protect academic performance and minimize dependency risks.

2. Adopt Sleep Hygiene & Ergonomic Break Protocols

Enforce a strict "No-Screen Buffer Hour" before bedtime to ensure better sleep quality, alongside practicing the 20-20-20 rule during active gaming sessions to reduce physical strain and eye fatigue.

3. Foster Offline Social Integration

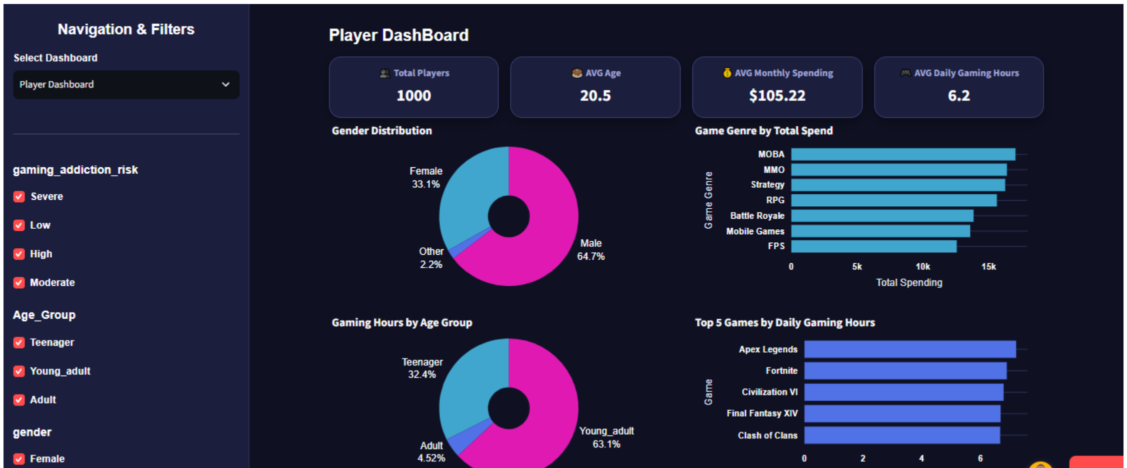
Universities and local communities should create hybrid or physical social clubs that align with esports interests, directly helping high-risk gamers convert online screen time into real-world, face-to-face social connections.

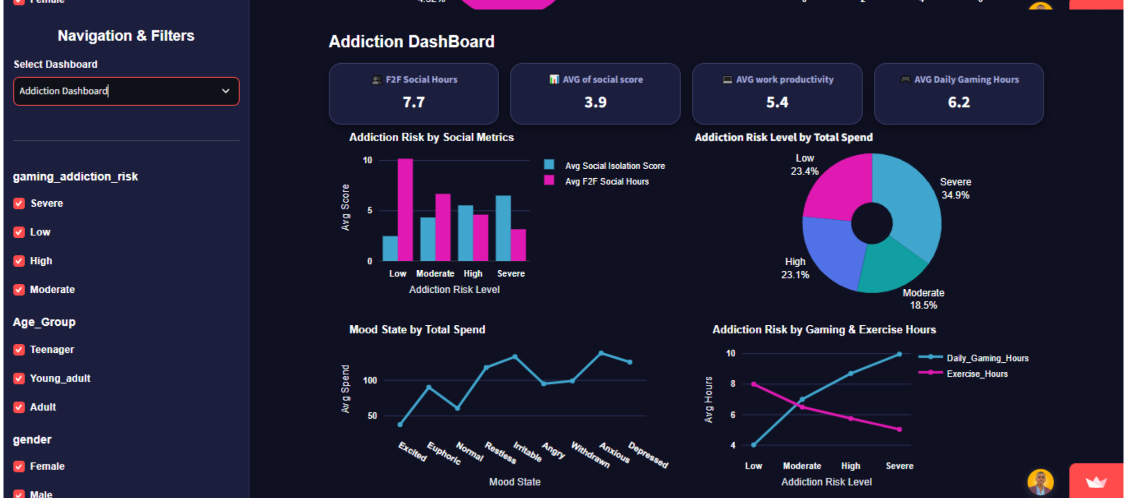
4. Establish Automated Micro-transaction Caps

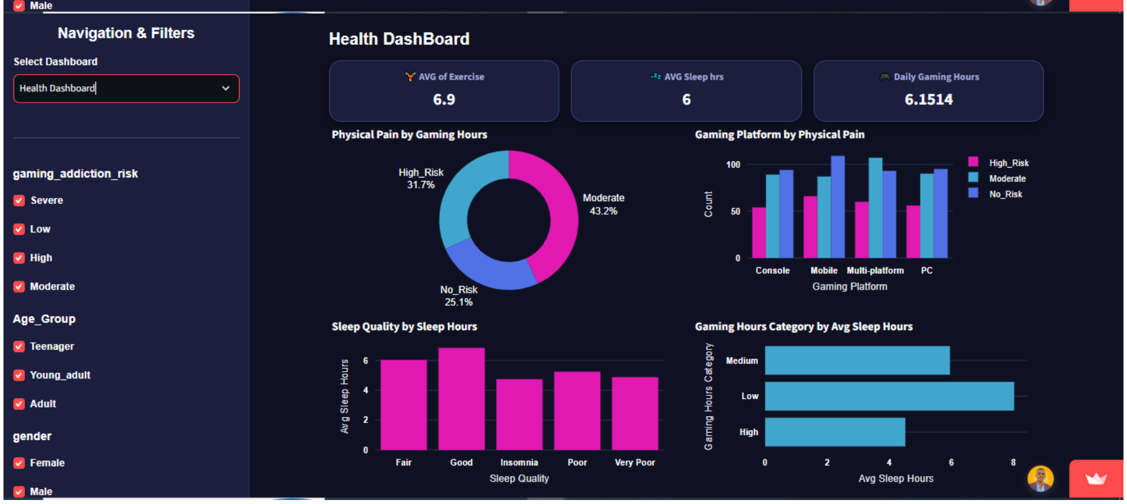
Players experiencing high impulse spending should set up automated, monthly micro-transaction boundaries directly inside their banking or payment applications, to avoid gaming-induced financial stress.

Summary

This document covers the full project journey: understanding the data and its challenges, descriptive and inferential statistics, data preprocessing and building a structured relational database (Star Schema), extracting KPIs and analyzing relationships between variables, and finally three interactive dashboards along with the final actionable recommendations.


Navigation & Filters
Select Dashboard
Player Dashboard
gaming_addiction_risk
Severe
Low
High
Moderate
Age_Group
Teenager
Young_adult
Adult
gender
Female
Player DashBoard
Total Players
1000
AVG Age
20.5
AVG Monthly Spending
$105.22
AVG Daily Gaming Hours
6.2
Gender Distribution
Female 33.1%
Other 2.2%
Male 64.7%
Game Genre by Total Spend
MOBA
MMO
Strategy
RPG
Battle Royale
Mobile Games
FPS
Game Genre
0
5k
10k
15k
Total Spending
Gaming Hours by Age Group
Teenager 32.4%
Adult 4.52%
Young_adult 63.1%
Top 5 Games by Daily Gaming Hours
Apex Legends
Fortnite
Civilization VI
Final Fantasy XIV
Clash of Clans
Game
0
2
4
6


Navigation & Filters
Select Dashboard
Addiction Dashboard
gaming_addiction_risk
Severe
Low
High
Moderate
Age_Group
Teenager
Young_adult
Adult
gender
Female
Male
Addiction DashBoard
F2F Social Hours
7.7
AVG of social score
3.9
AVG work productivity
5.4
AVG Daily Gaming Hours
6.2
Addiction Risk by Social Metrics
Avg Social Isolation Score
Avg F2F Social Hours
Avg Score
10
5
0
Low
Moderate
High
Severe
Addiction Risk Level
Addiction Risk Level by Total Spend
Low 23.4%
Severe 34.9%
High 23.1%
Moderate 18.5%
Mood State by Total Spend
Avg Spend
100
50
Excited
Euphoric
Normal
Restless
Irritable
Angry
Withdrawn
Anxious
Depressed
Mood State
Addiction Risk by Gaming & Exercise Hours
Daily_Gaming_Hours
Exercise_Hours
Avg Hours
10
8
6
4
Low
Moderate
High
Severe
Addiction Risk Level


Navigation & Filters
Select Dashboard
Health Dashboard
gaming_addiction_risk
Severe
Low
High
Moderate
Age_Group
Teenager
Young_adult
Adult
gender
Female
Male
Health DashBoard
AVG of Exercise
6.9
AVG Sleep hrs
6
Daily Gaming Hours
6.1514
Physical Pain by Gaming Hours
High_Risk 31.7%
Moderate 43.2%
No_Risk 25.1%
Gaming Platform by Physical Pain
High_Risk
Moderate
No_Risk
Count
100
50
0
Console
Mobile
Multi-platform
PC
Gaming Platform
Sleep Quality by Sleep Hours
Avg Sleep Hours
6
4
2
0
Fair
Good
Insomnia
Poor
Very Poor
Sleep Quality
Gaming Hours Category by Avg Sleep Hours
Medium
Low
High
Gaming Hours Category
0
2
4
6
8
Avg Sleep Hours

18. Conclusion and Future Work

18.1 Conclusion

This project successfully analyzed the relationship between gaming behavior and mental health using Excel, SQL, Python, Power BI, and Tableau. Through data preprocessing, exploratory analysis, and dashboard development, meaningful insights were generated to better understand how gaming habits may influence mental well-being.

18.2 Future Work

Future improvements may include applying machine learning models to predict addiction risk, integrating larger datasets, and developing real-time dashboards to support continuous analysis and decision-making.

References

 Coolors. (n.d.). *Color Palette Generator*. Retrieved from https://coolors.co/

 Flaticon. (n.d.). *Free Vector Icons*. Retrieved from https://www.flaticon.com/

 Kaggle. (n.d.). *Gaming and Mental Health Dataset*. Retrieved from https://www.kaggle.com/datasets/shaistashahid/gaming-and-mental-health